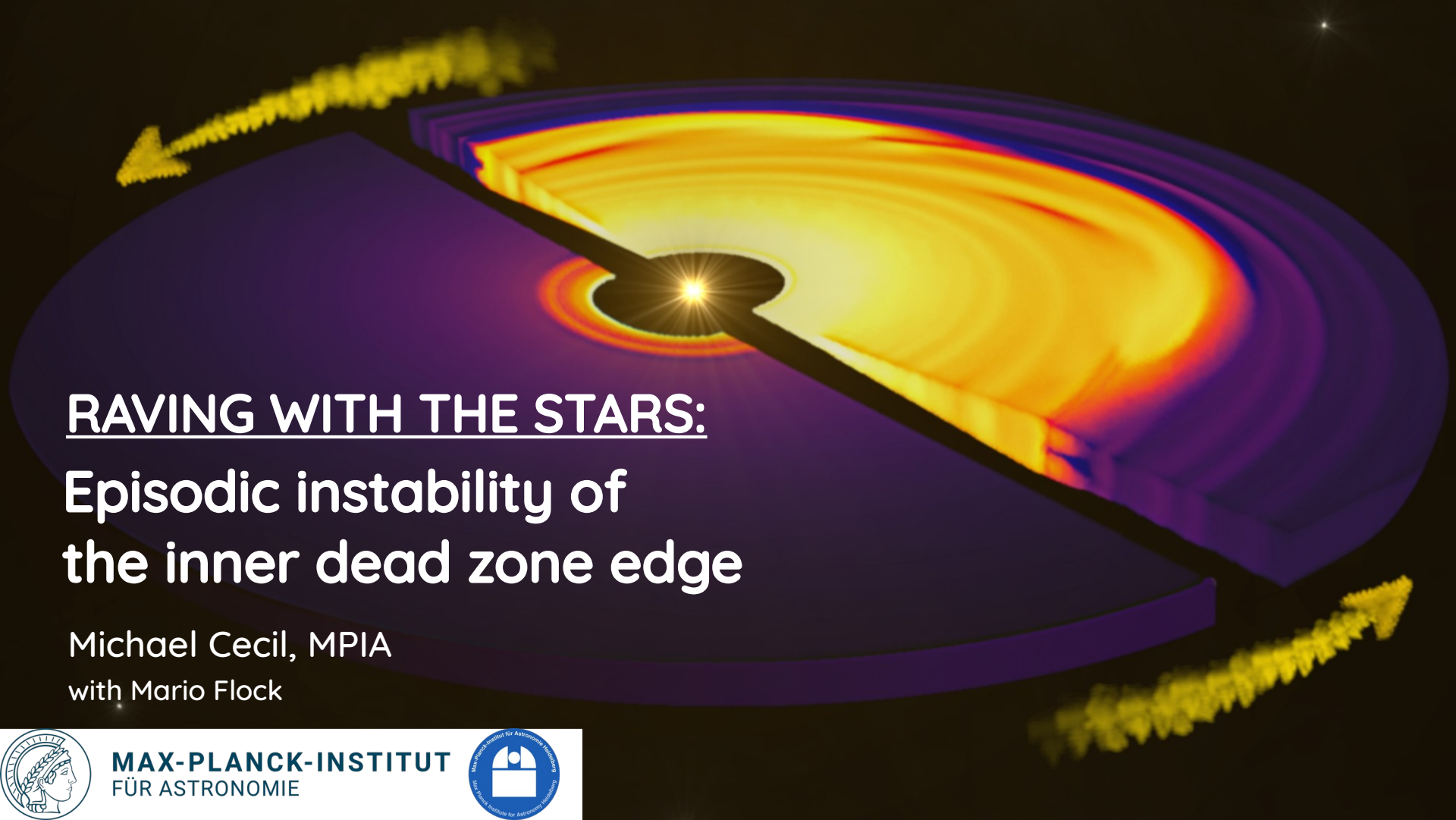




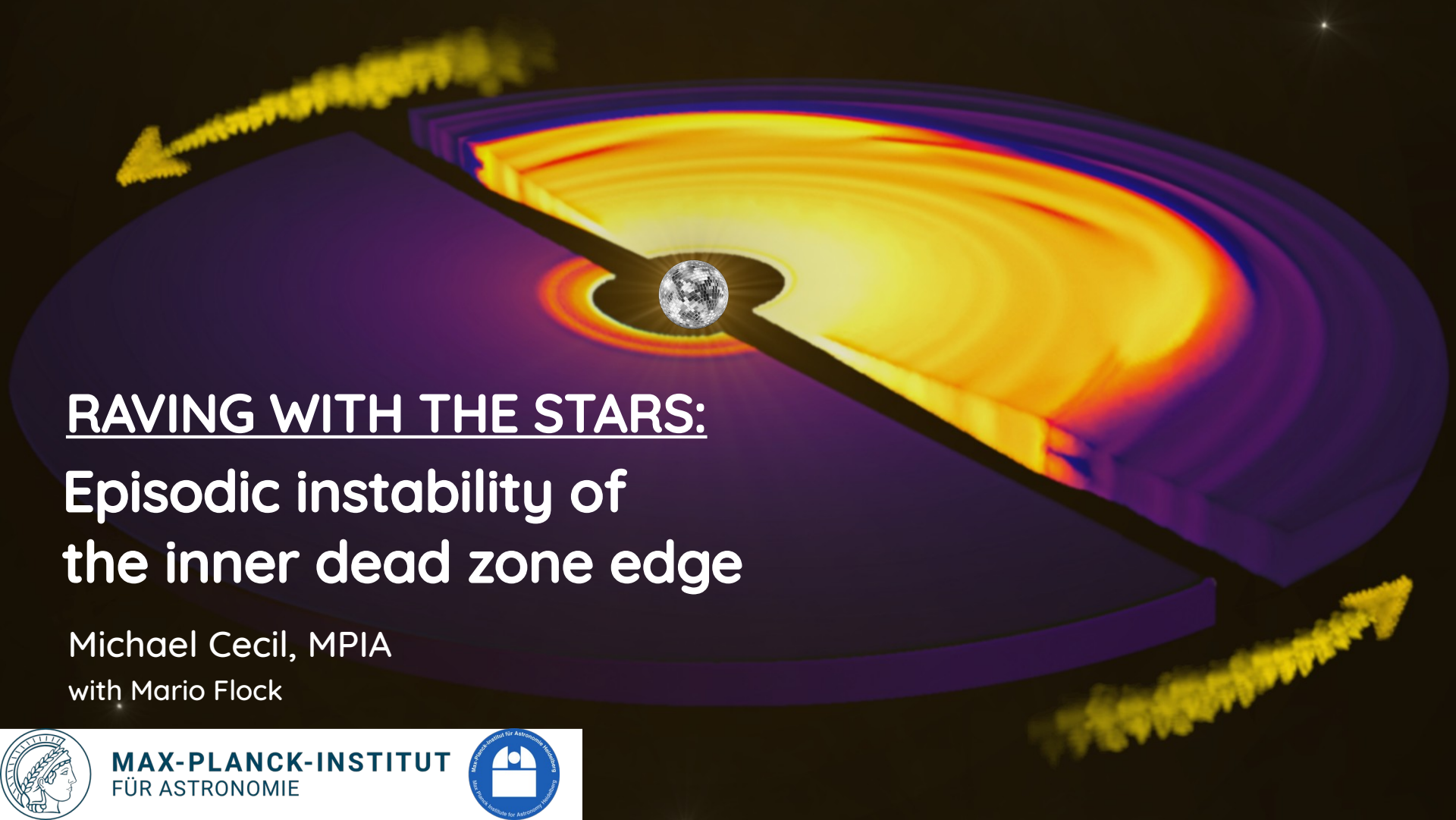
RAVING WITH THE STARS: Episodic instability of the inner dead zone edge

Michael Cecil, MPA
with Mario Flock



MAX-PLANCK-INSTITUT
FÜR ASTRONOMIE





RAVING WITH THE STARS: Episodic instability of the inner dead zone edge

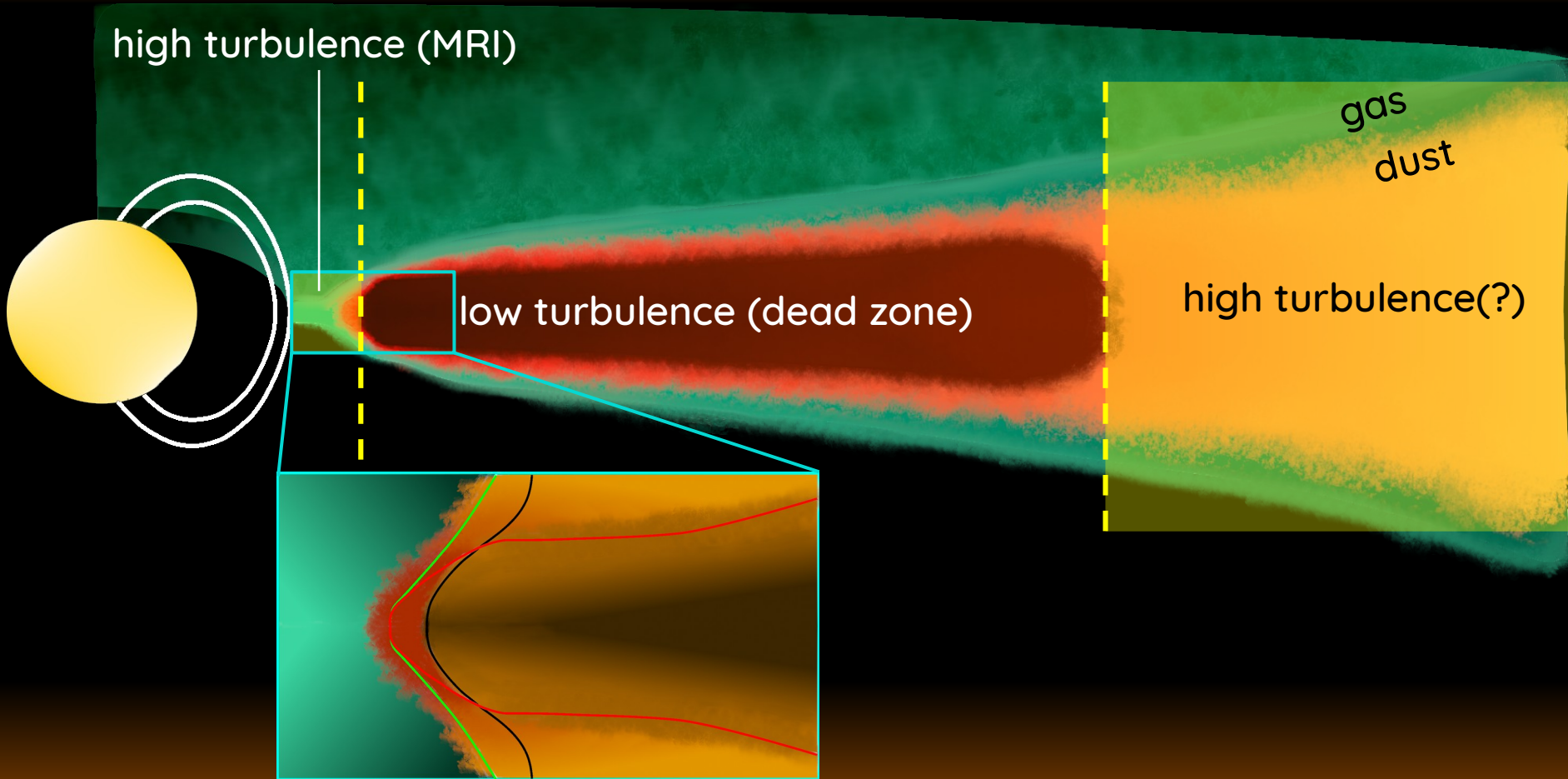
Michael Cecil, MPIA
with Mario Flock



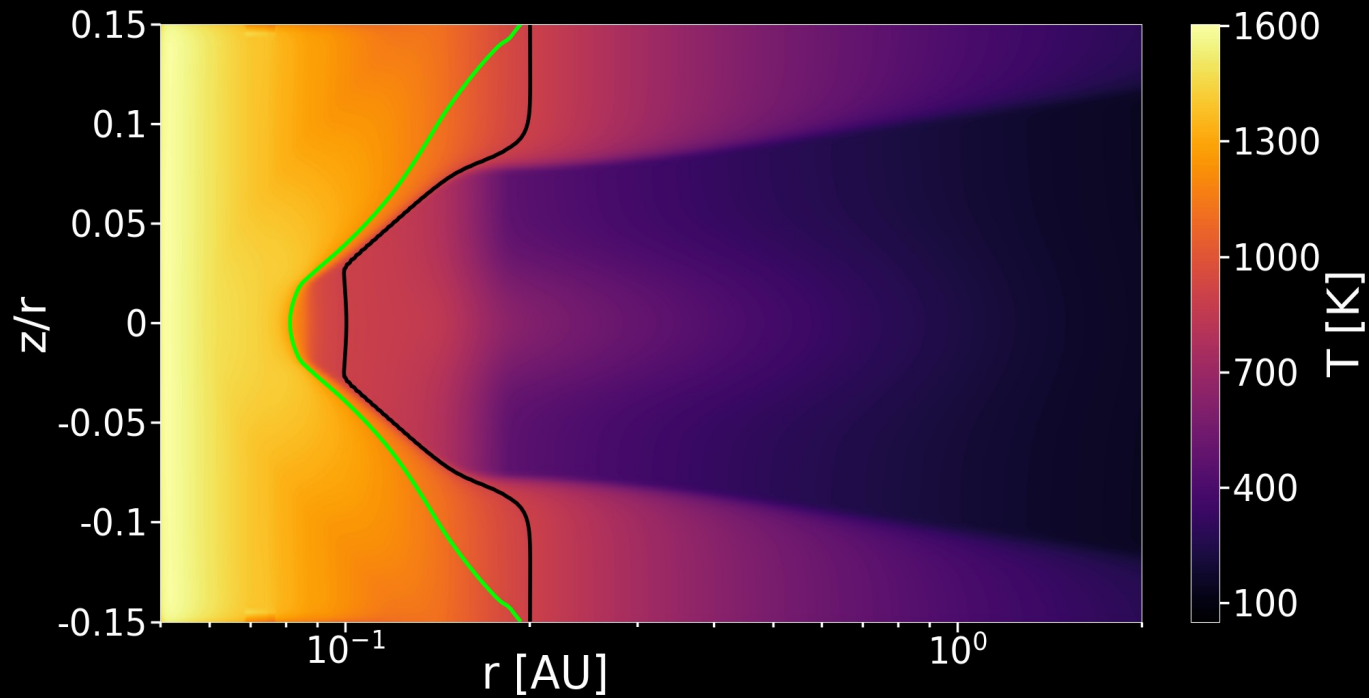
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Disk structure schematic

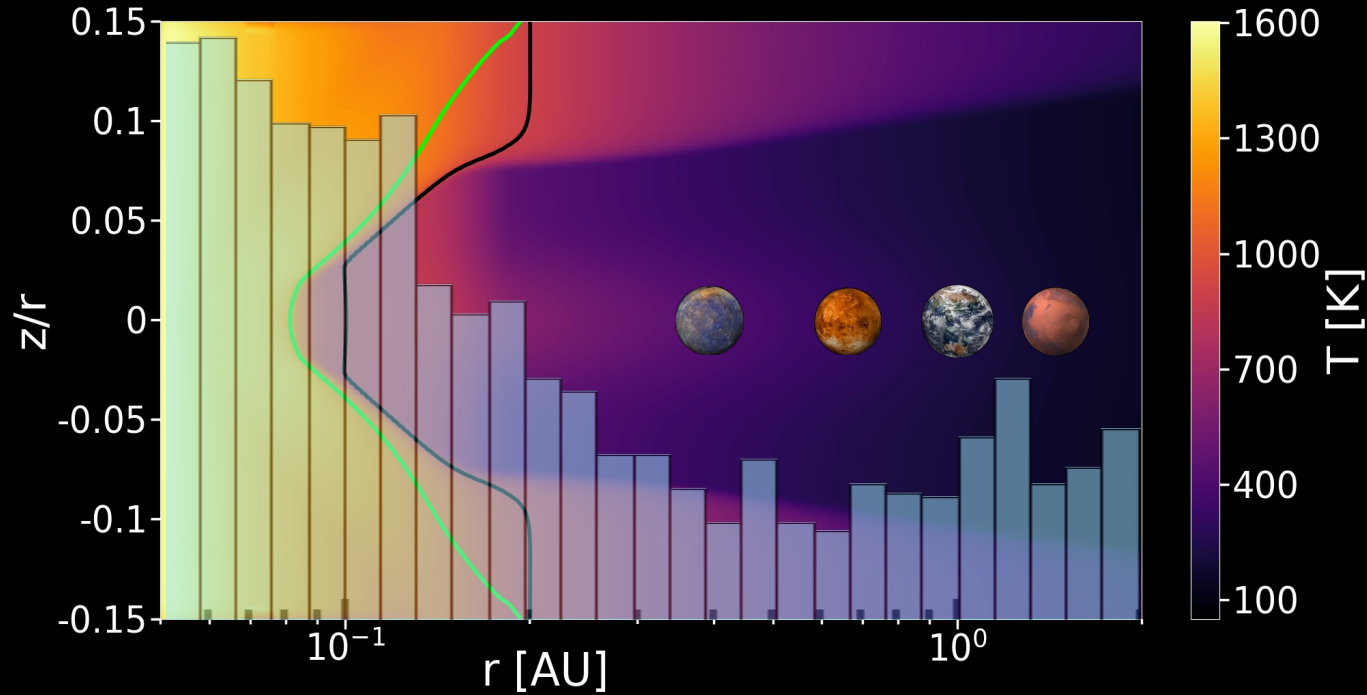


Radiation Hydrodynamics: Inner disk structure

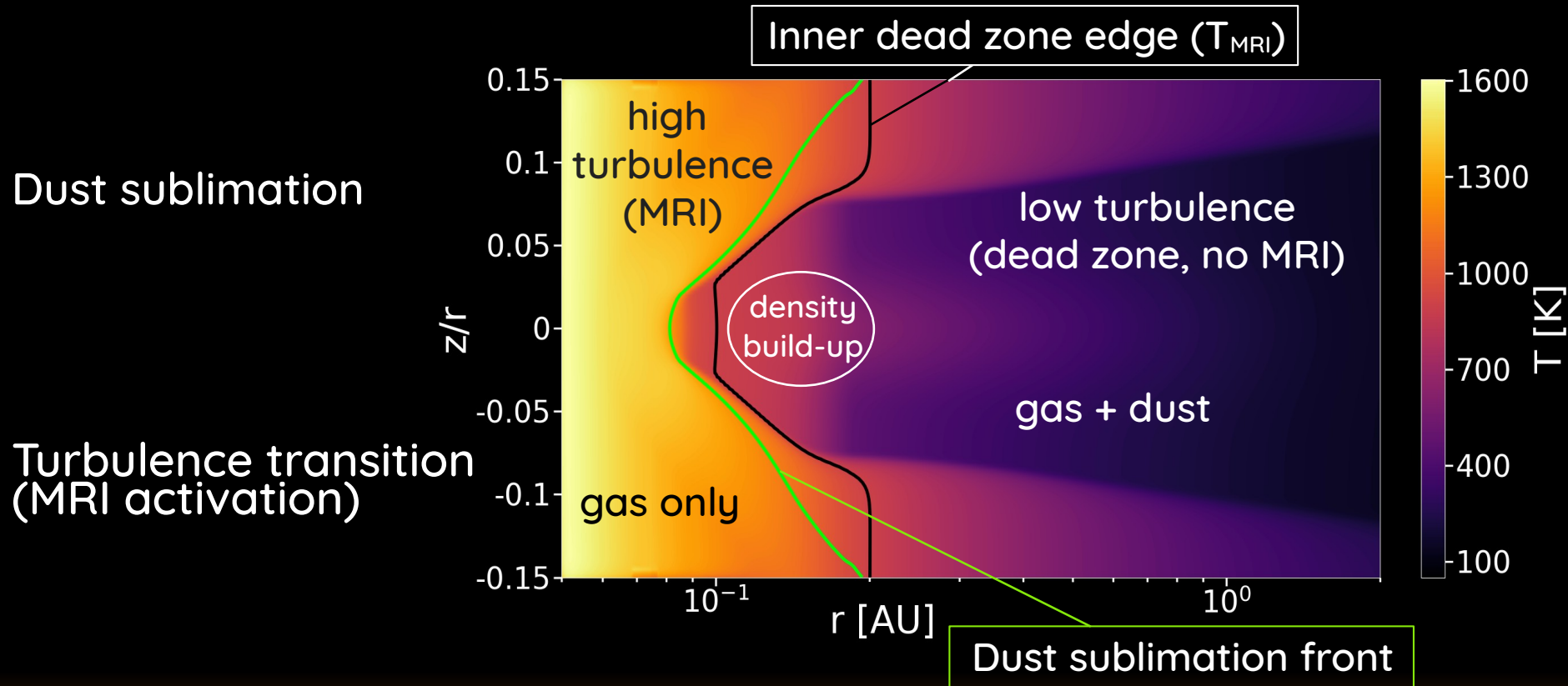


Radiation Hydrodynamics: Inner disk structure

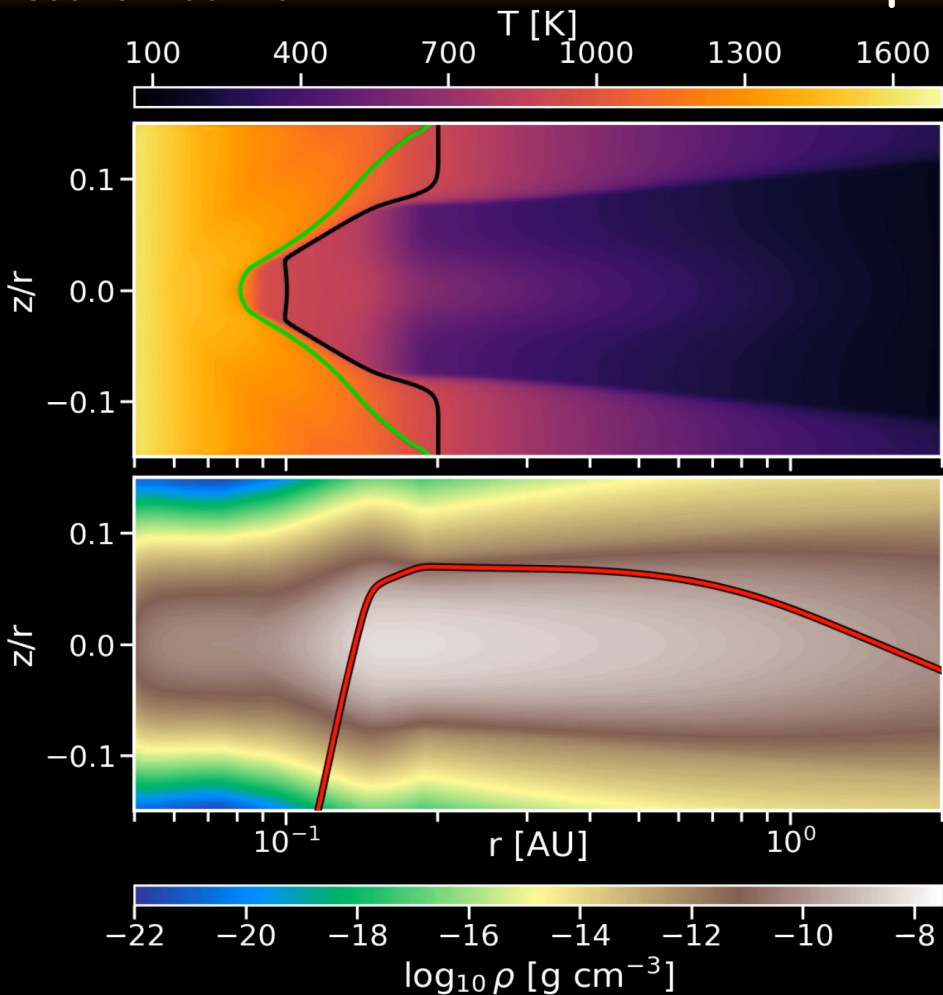
~85% of confirmed
exoplanets are at
 $r < 2$ AU



Radiation Hydrodynamics: Inner disk structure

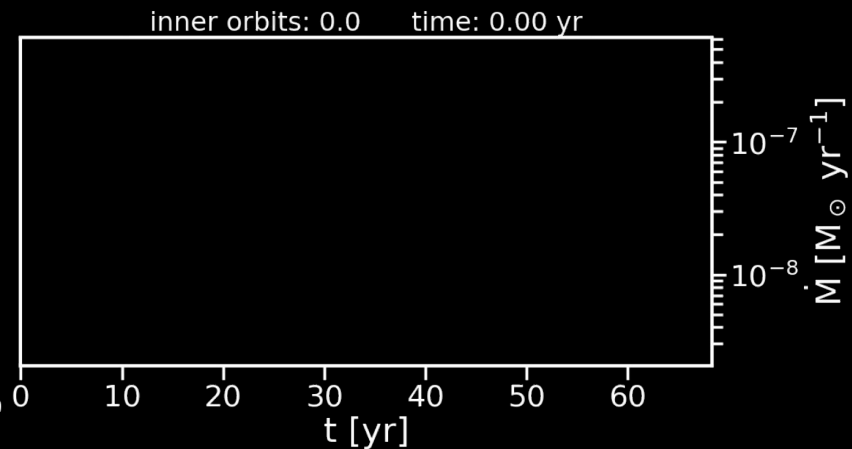


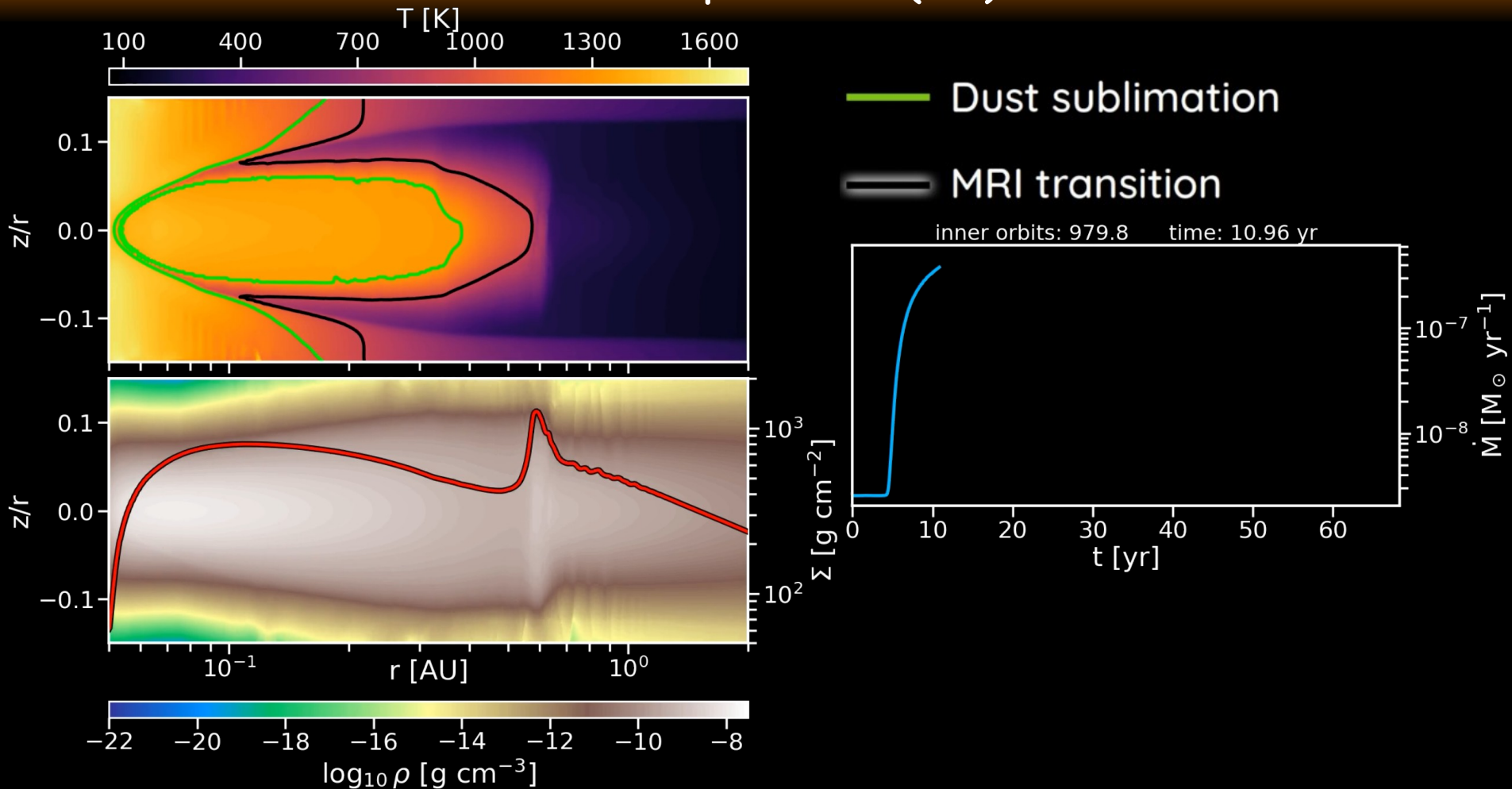
Is this long-term stable?

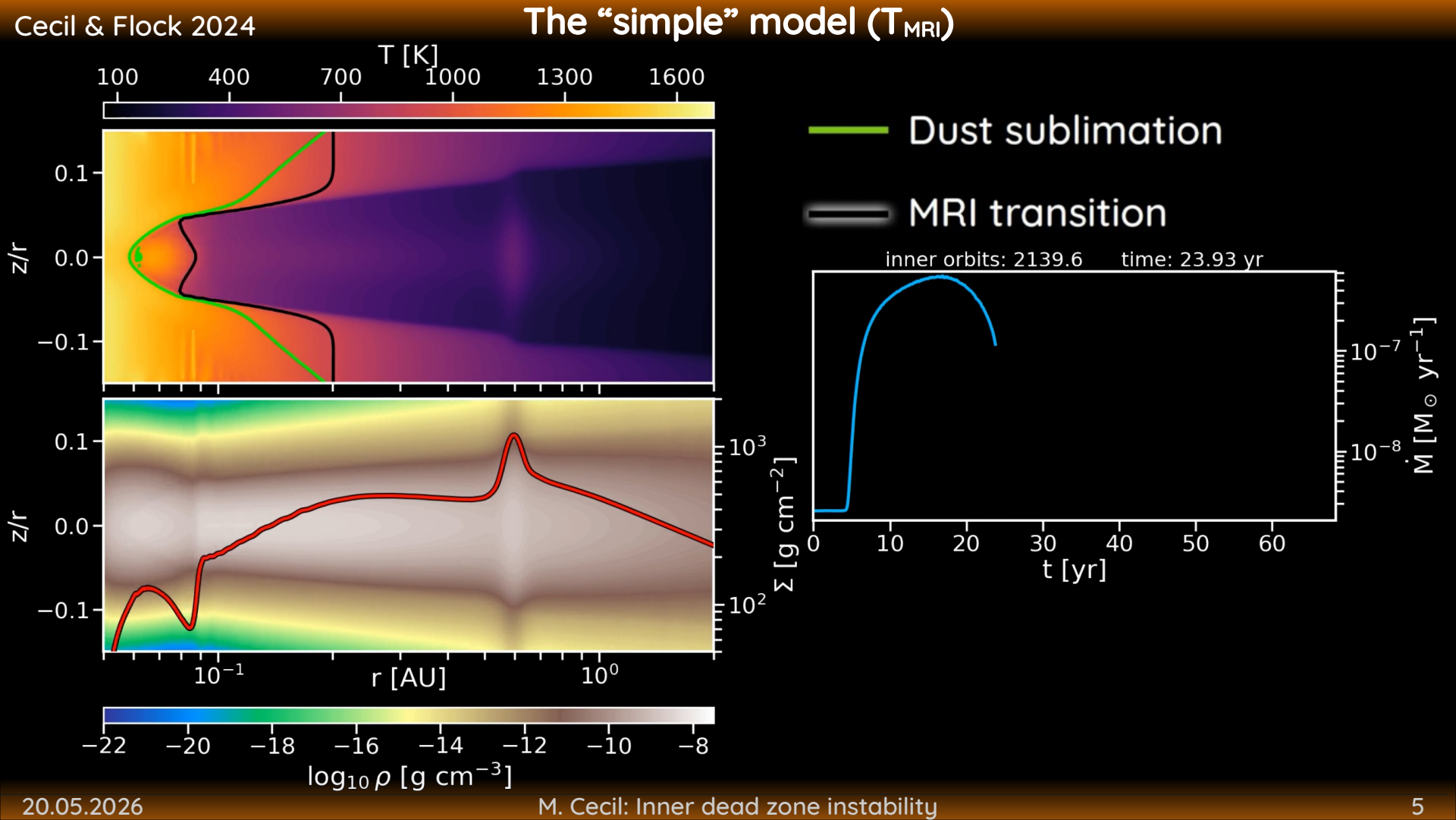


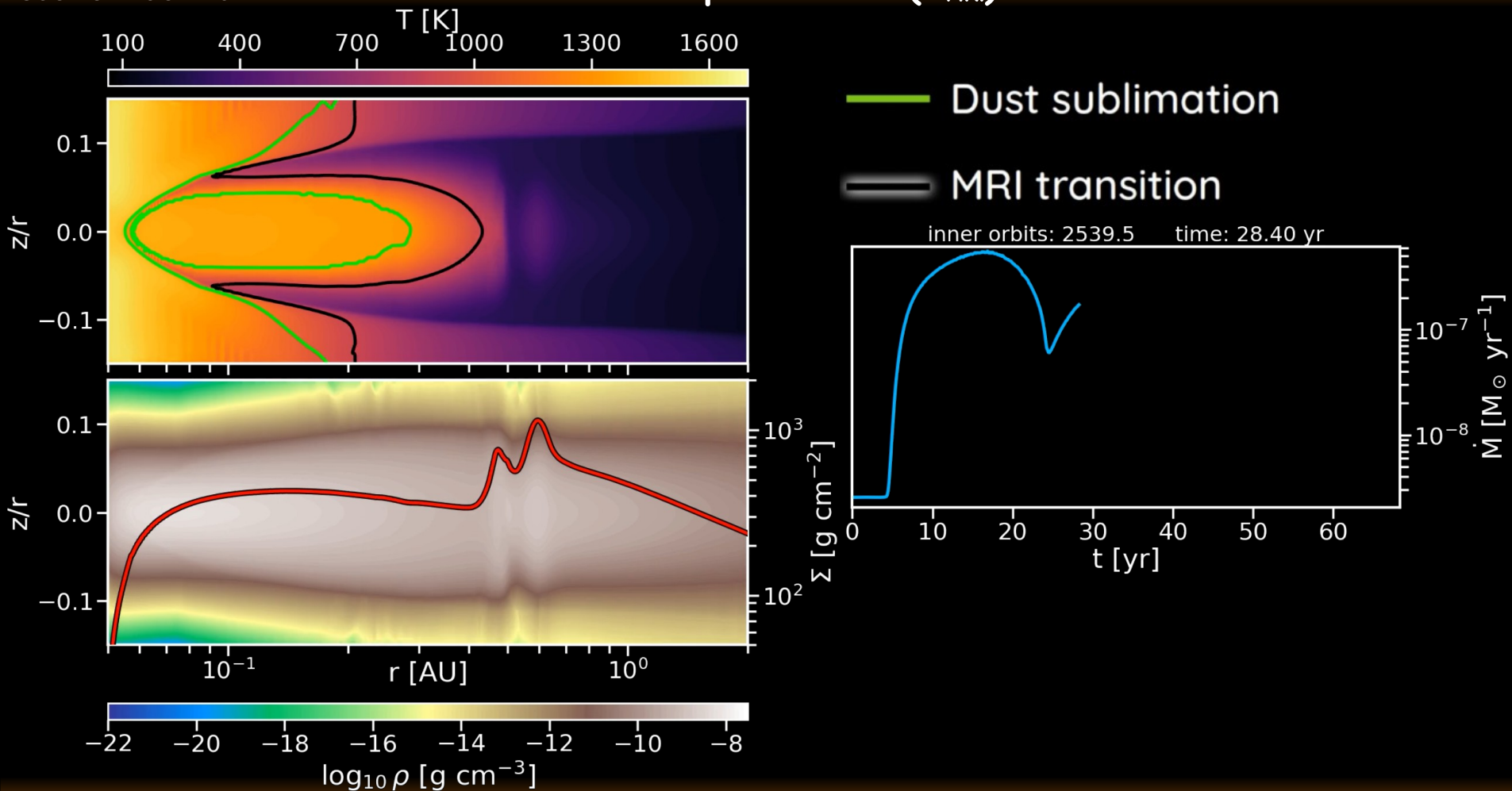
— Dust sublimation

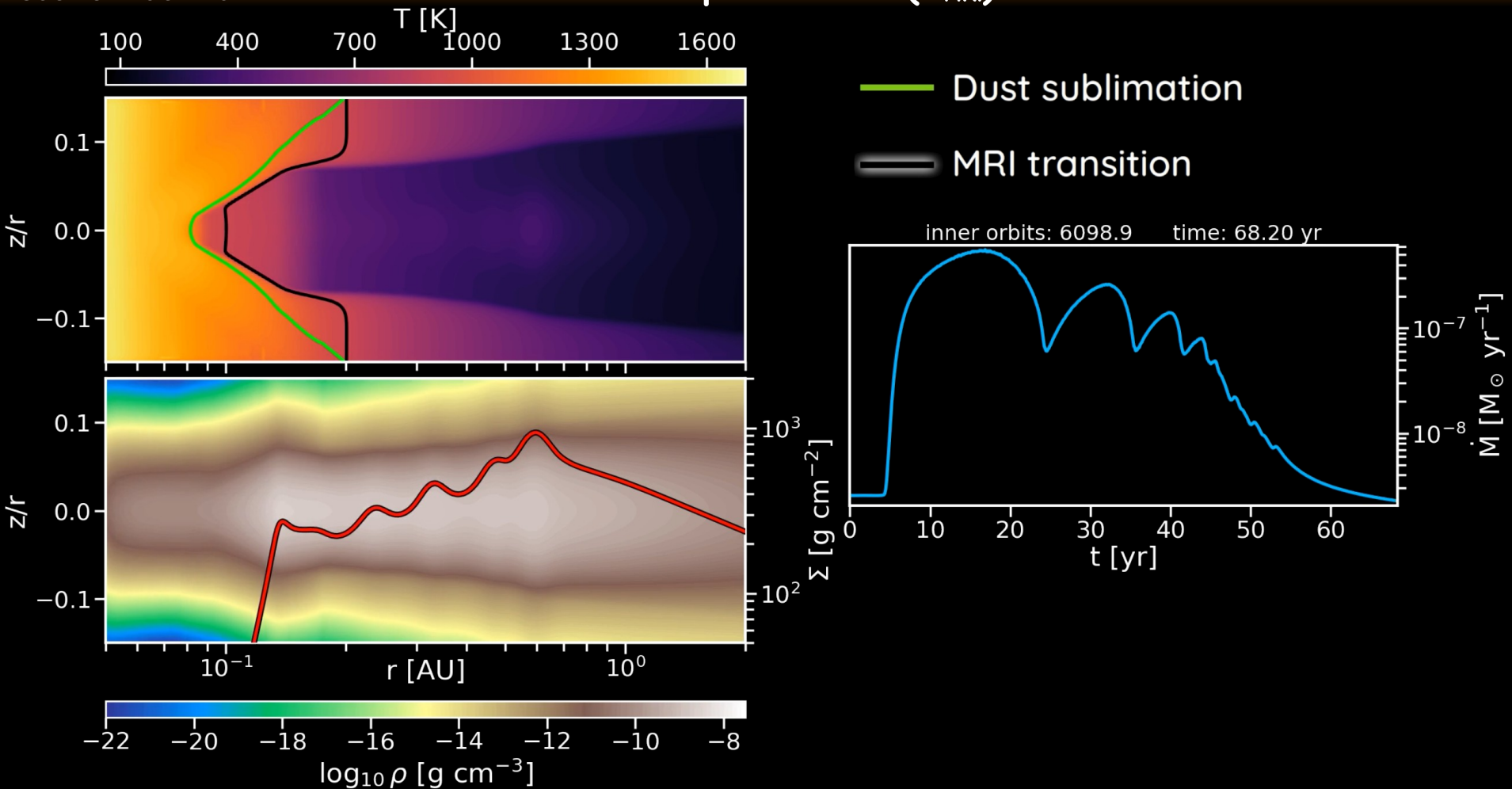
— MRI transition



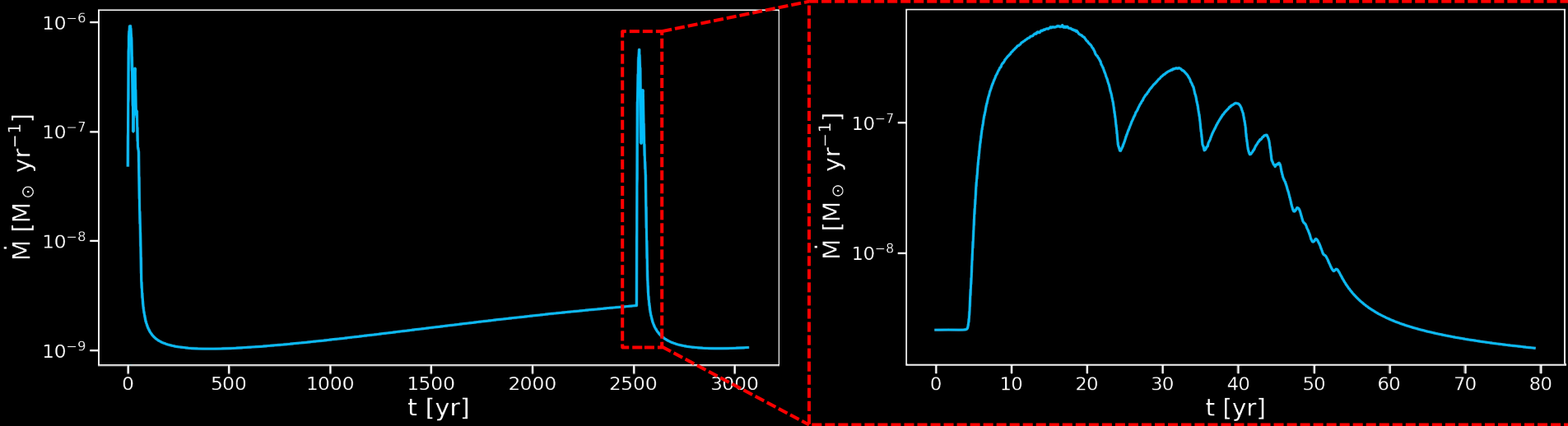




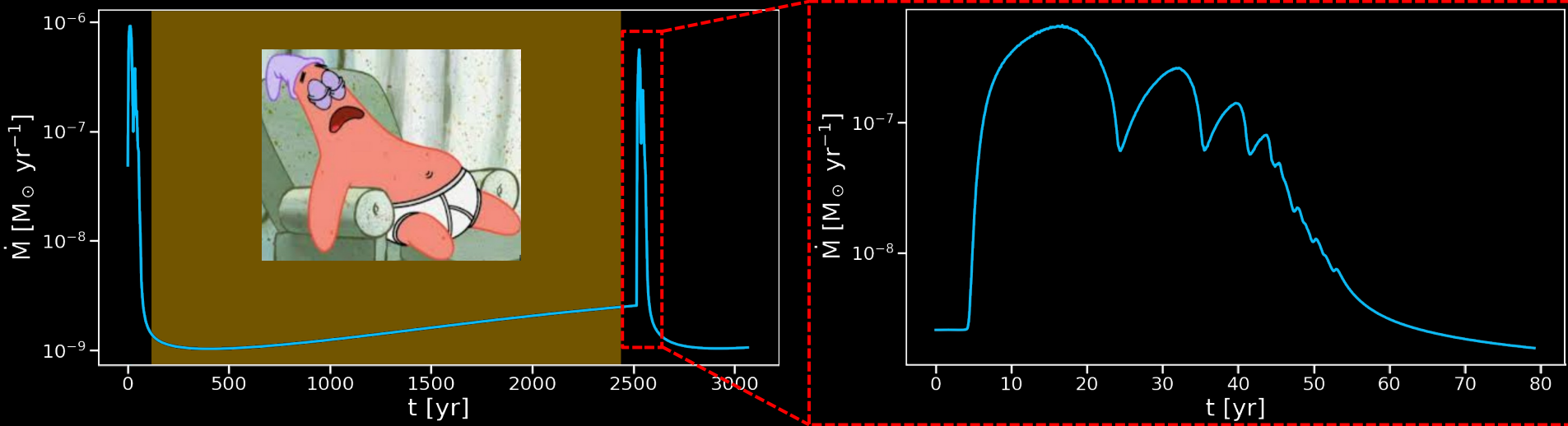


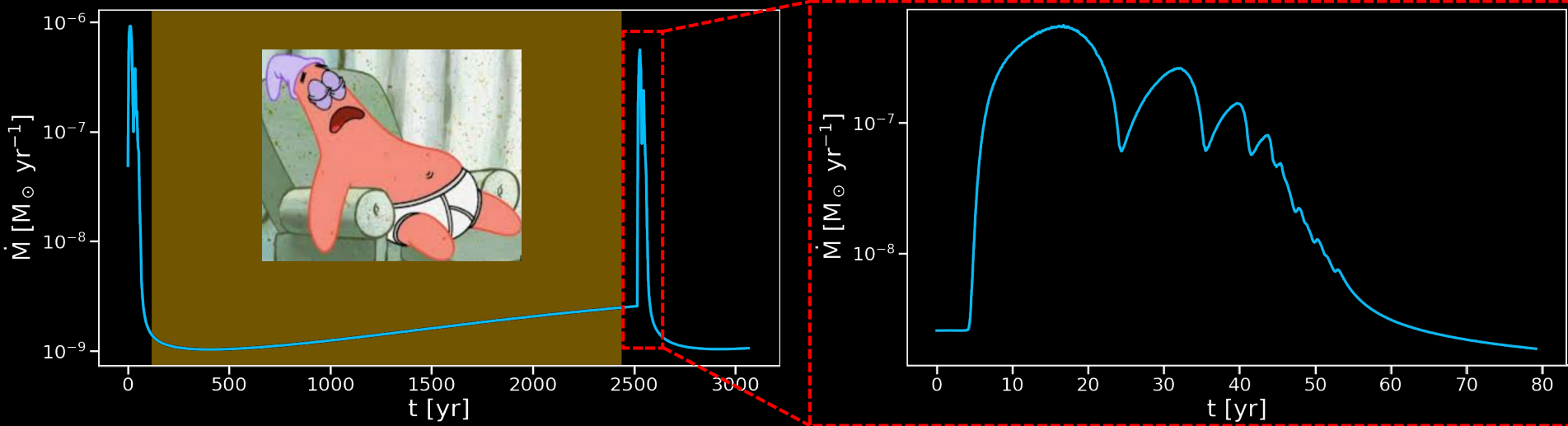


The “simple” model (T_{MRI}): Accretion rate modulation



The “simple” model (T_{MRI}): Accretion rate modulation





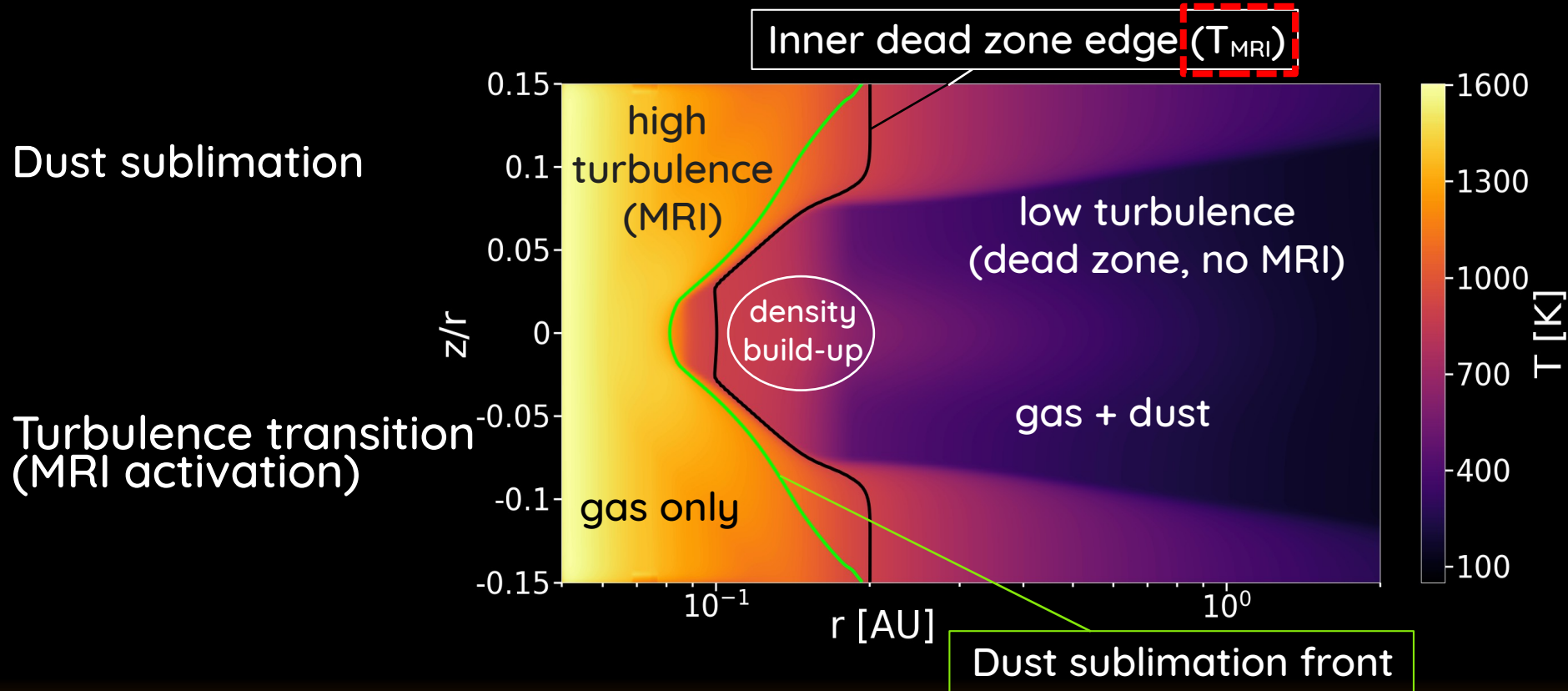
Inner dead zone instability produces episodic accretion outbursts

Burst timescales highly dependent on system properties

(e.g. mass, turbulence levels, luminosity, ...),

especially opacities (Cecil et al. 2026a)

Radiation Hydrodynamics: Inner disk structure

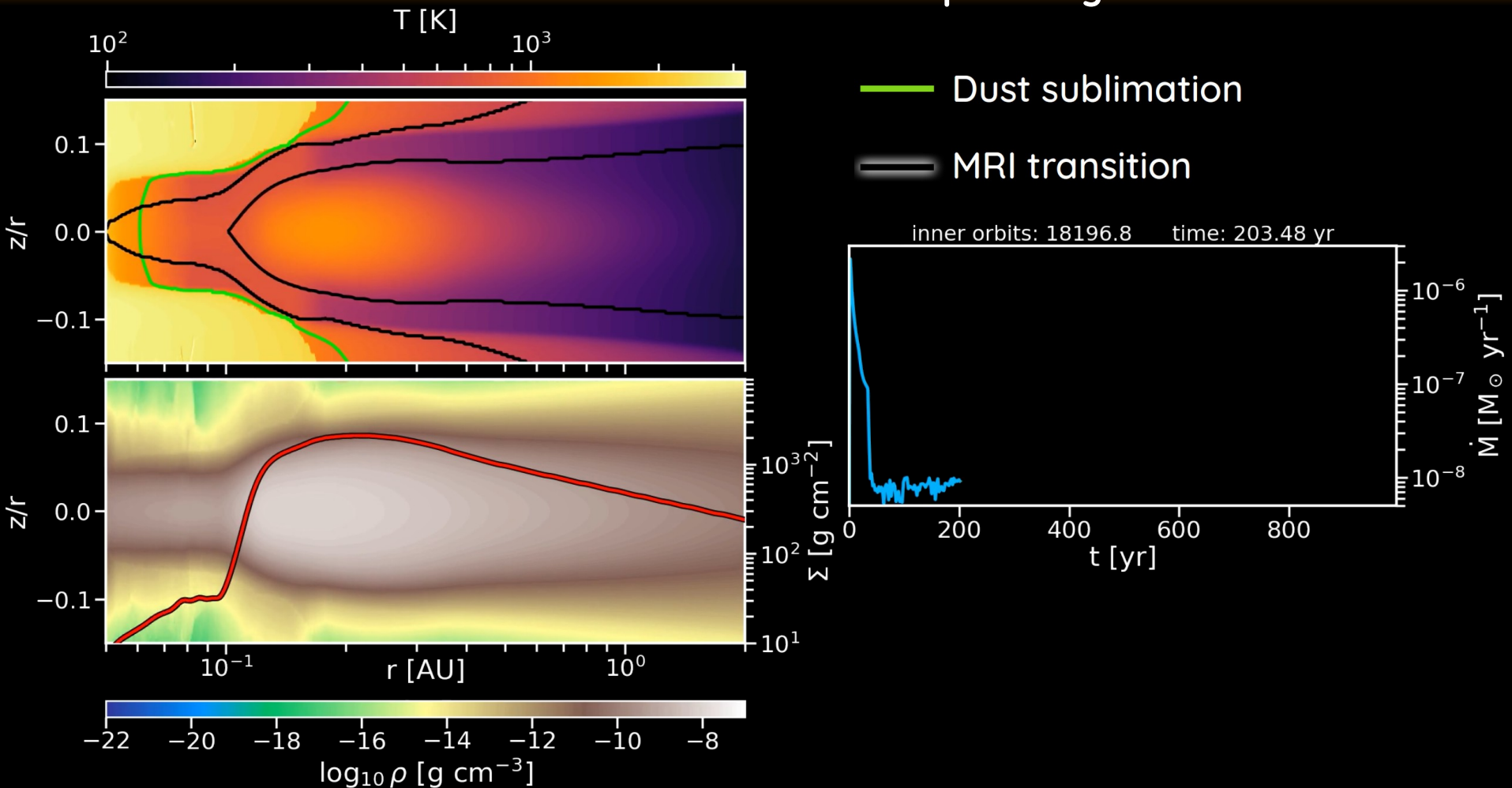


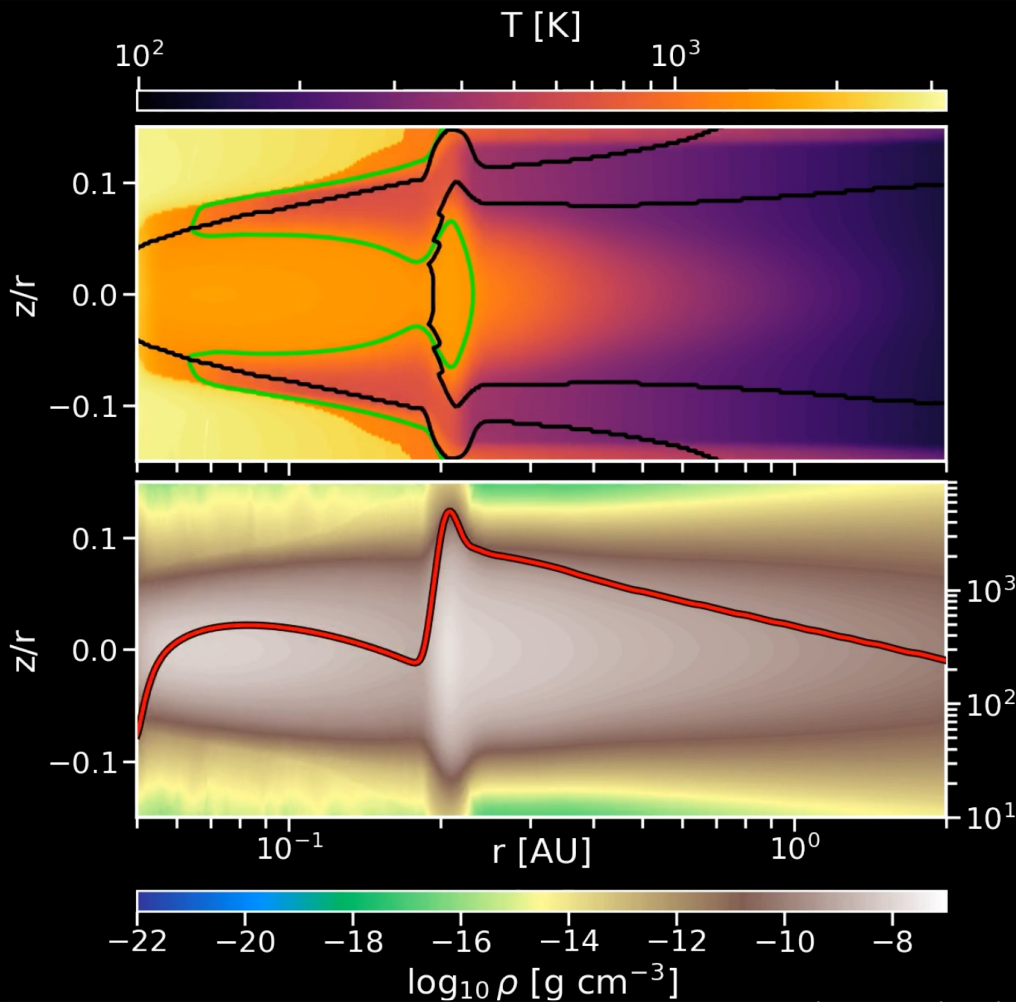
Locations of MRI activity:

What about magnetic fields???

non-ideal MHD?

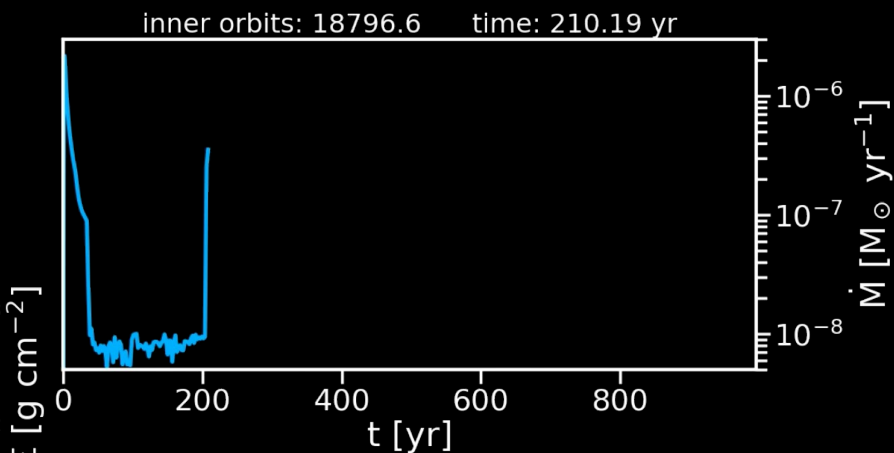
→ pseudo-non-ideal M-RHD

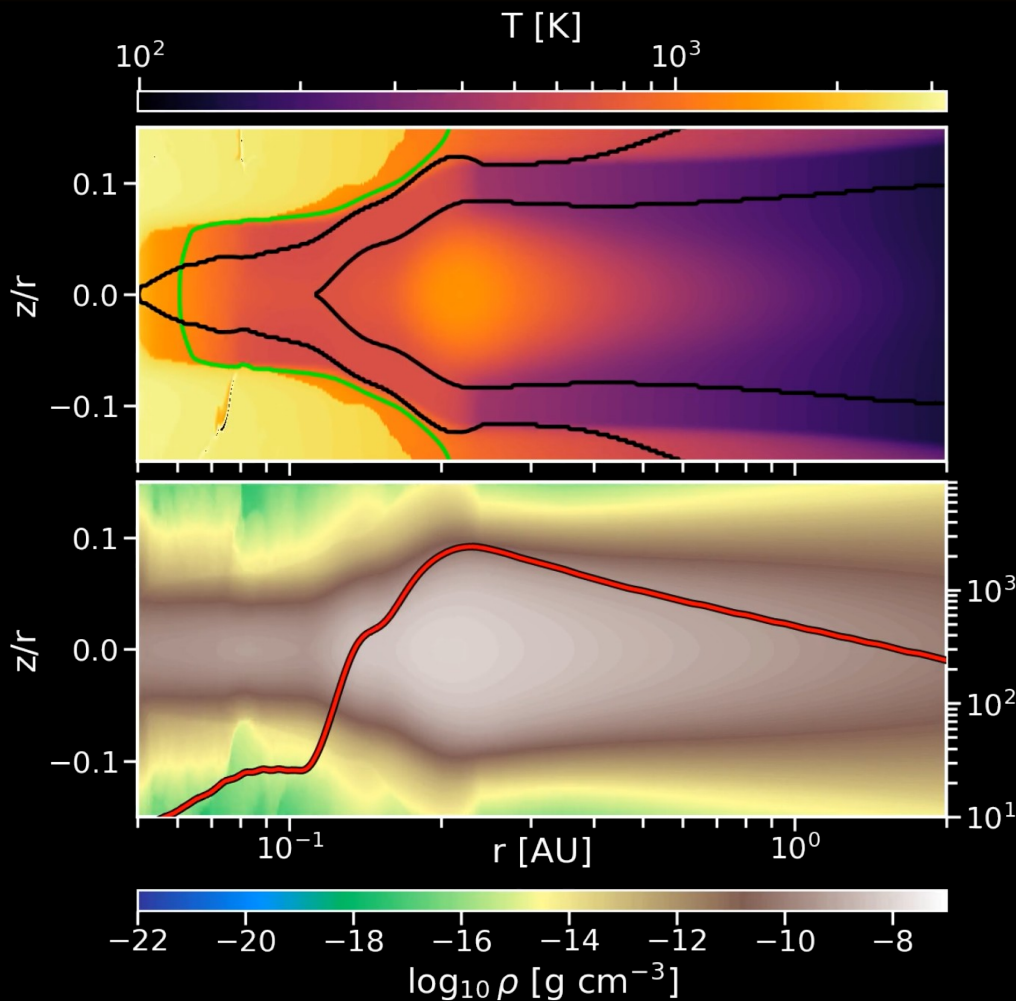




— Dust sublimation

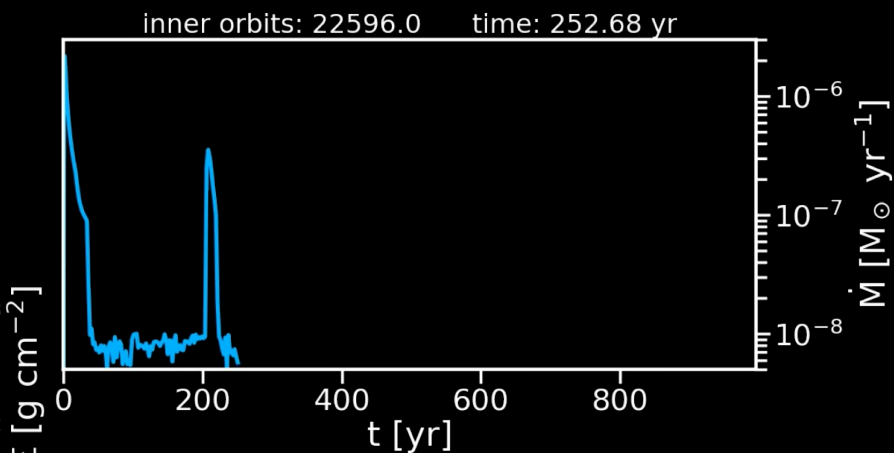
— MRI transition

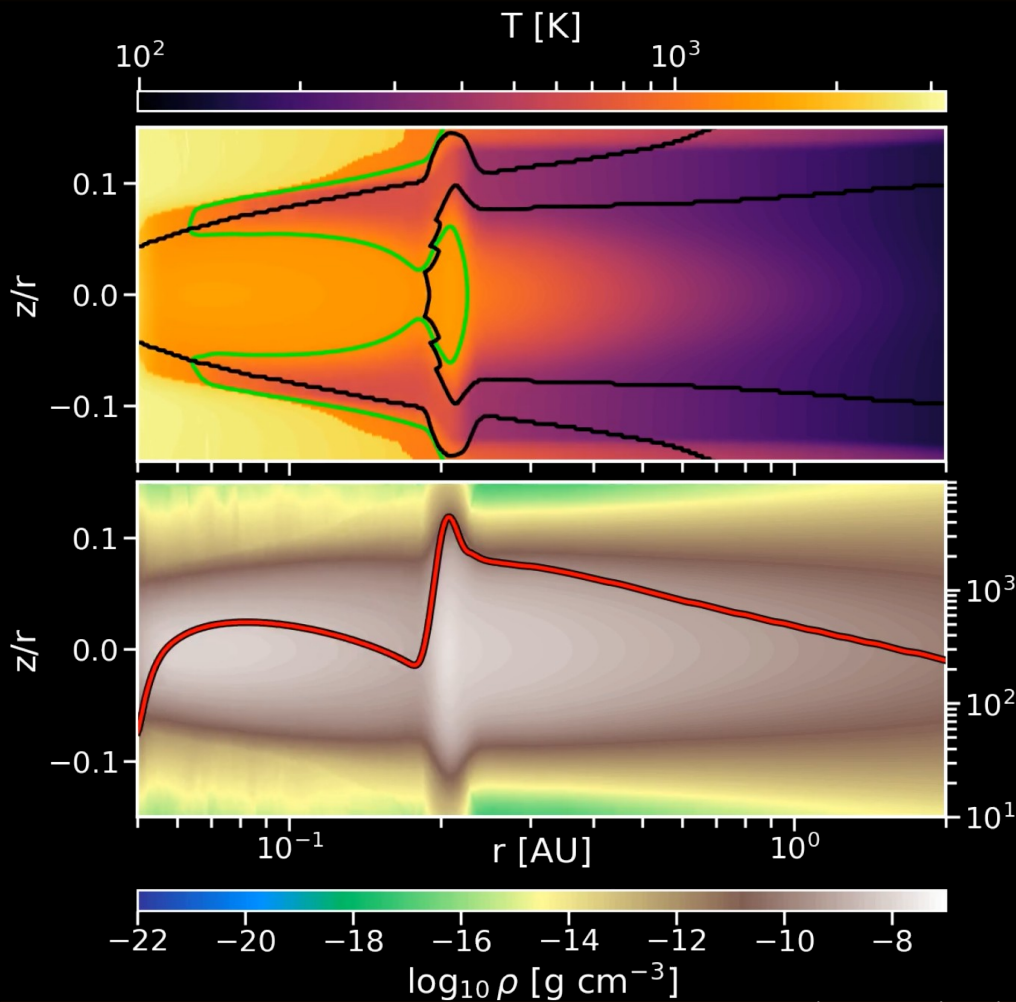




— Dust sublimation

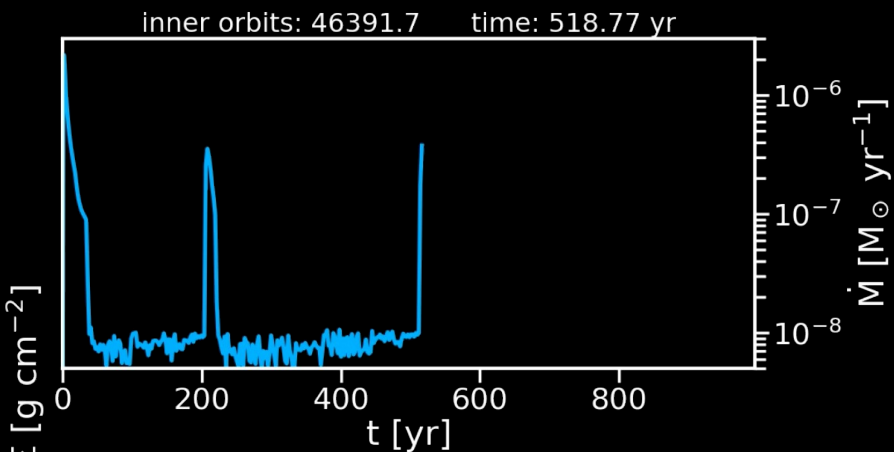
— MRI transition

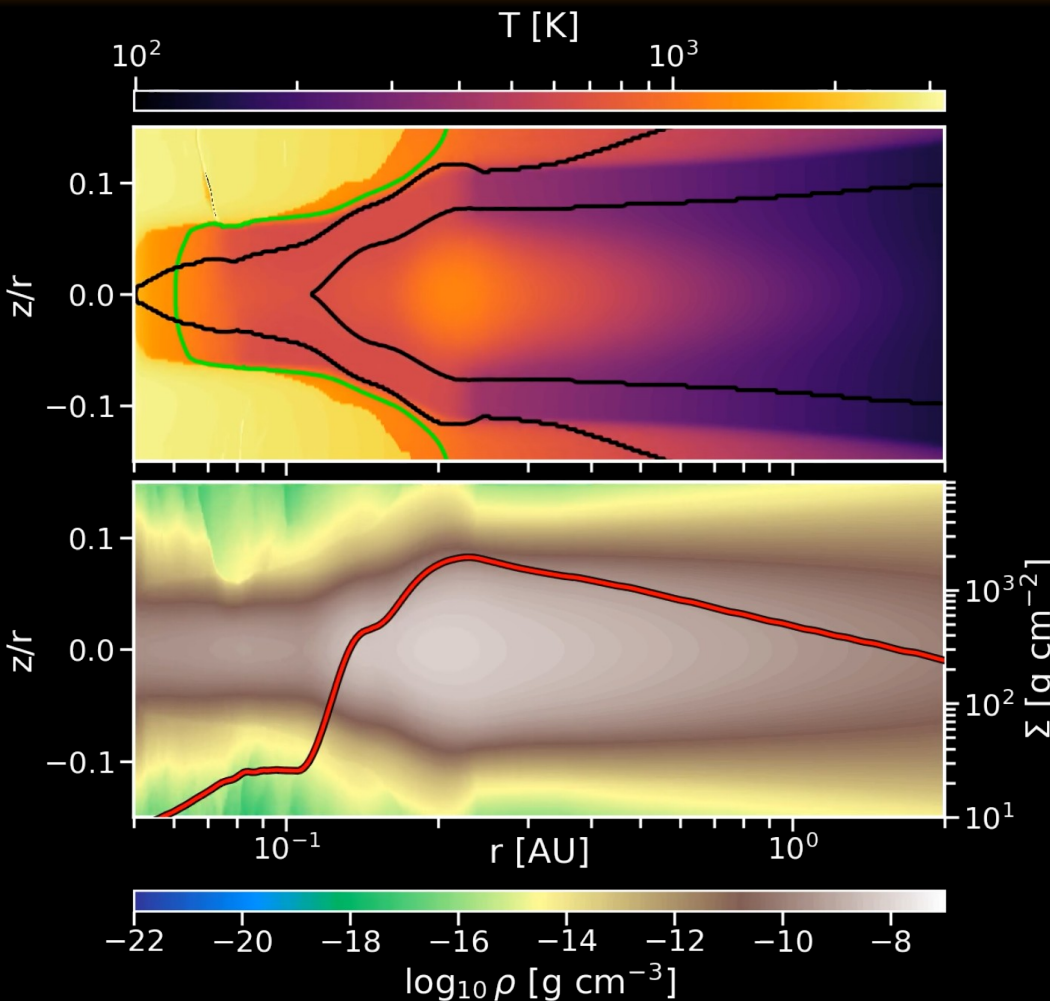




— Dust sublimation

— MRI transition

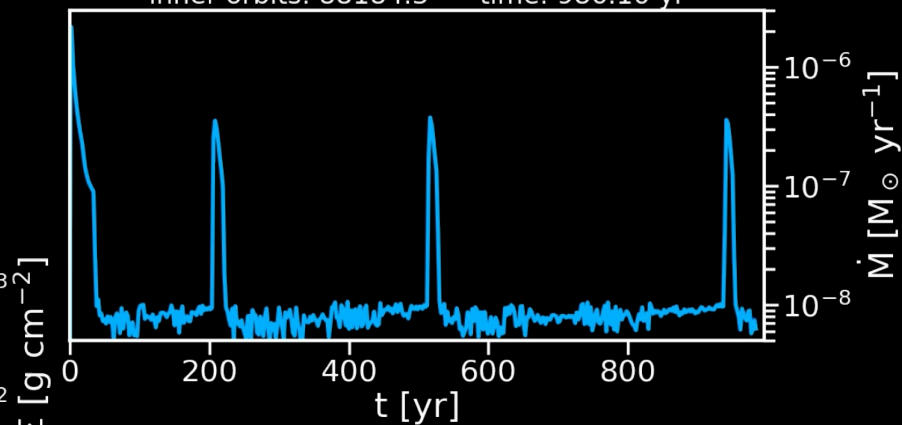




Dust sublimation

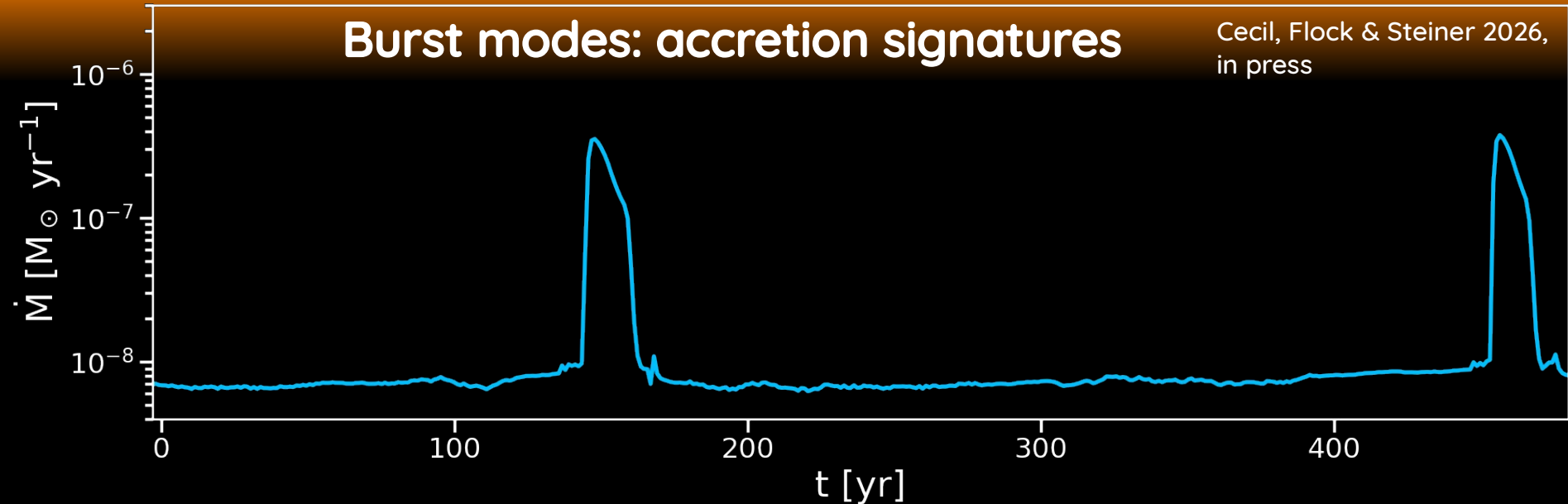
MRI transition

inner orbits: 88184.3 time: 986.10 yr



Burst modes: accretion signatures

Cecil, Flock & Steiner 2026,
in press



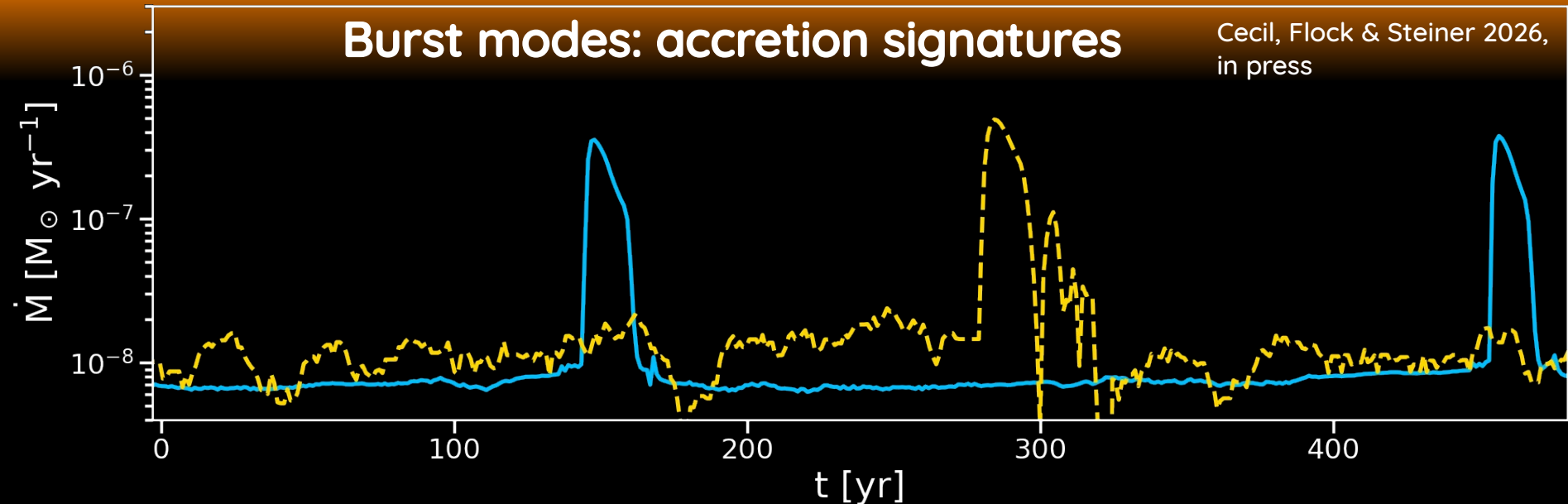
narrow burst mode

no reflare

— weak stellar field
— weak disk field

Burst modes: accretion signatures

Cecil, Flock & Steiner 2026,
in press



narrow burst mode

no reflare

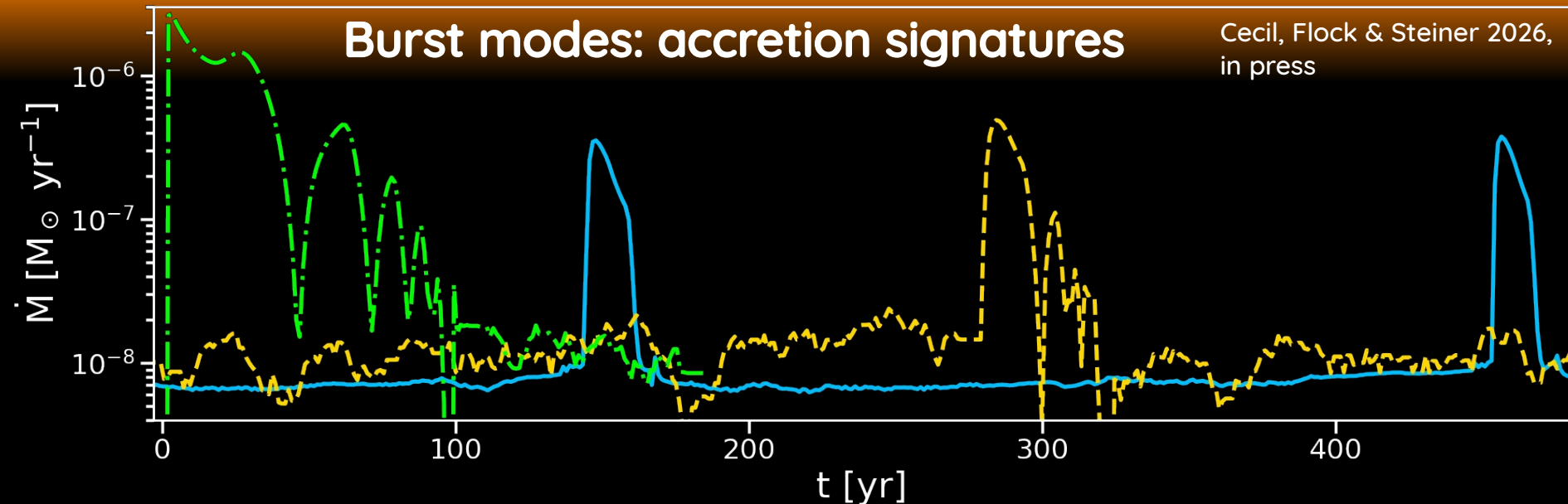
— weak stellar field
weak disk field

reflare

- - - strong stellar field
weak disk field

Burst modes: accretion signatures

Cecil, Flock & Steiner 2026,
in press



narrow burst mode

wide burst mode

no reflare

— weak stellar field
weak disk field

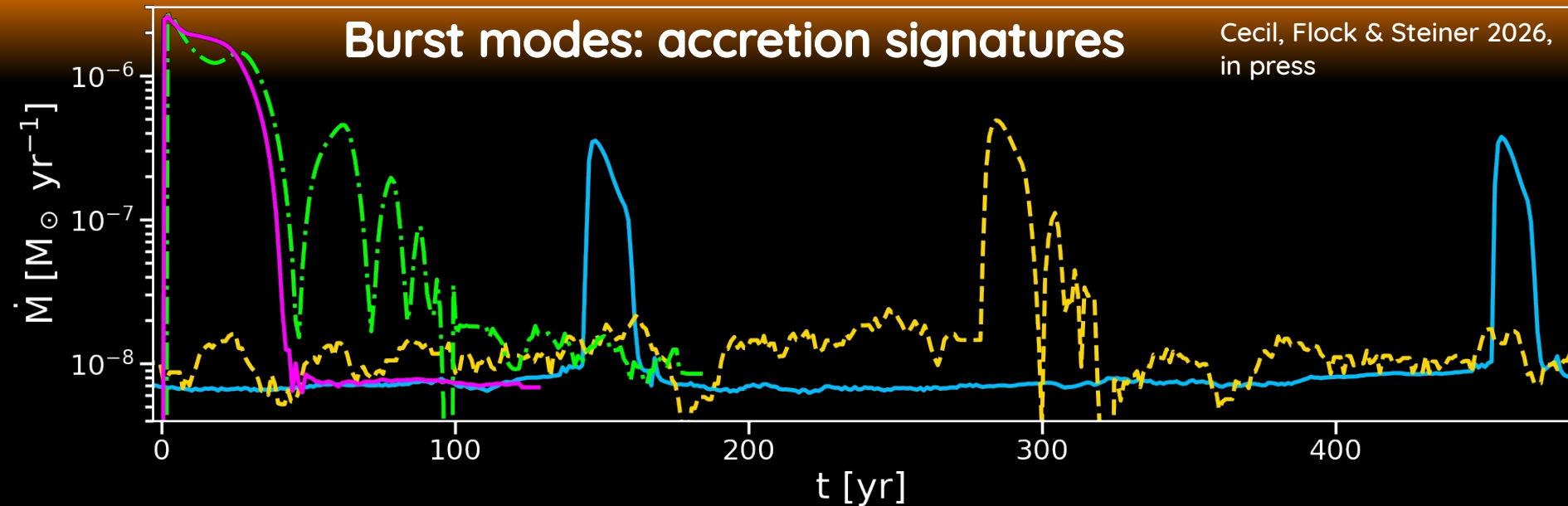
reflare

- - - strong stellar field
weak disk field

- . - strong stellar field
strong disk field

Burst modes: accretion signatures

Cecil, Flock & Steiner 2026,
in press



narrow burst mode

wide burst mode

no reflare

— weak stellar field
— weak disk field

— weak stellar field
— strong disk field

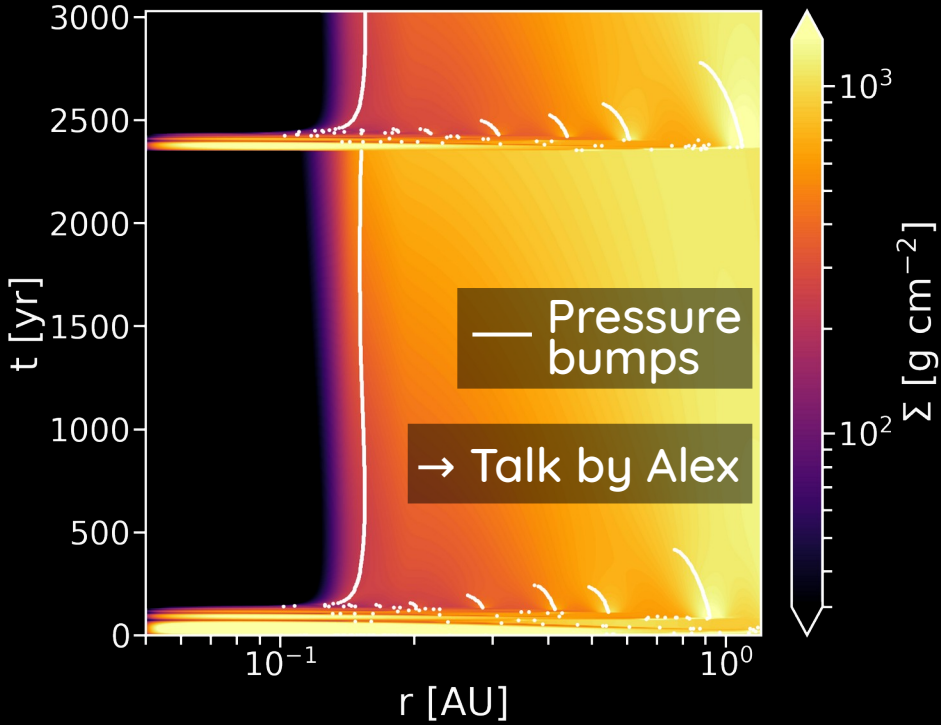
reflare

- - - strong stellar field
- - - weak disk field

- . - strong stellar field
- . - strong disk field

Wide burst mode

(equivalent “simple” T_{MRI} model)

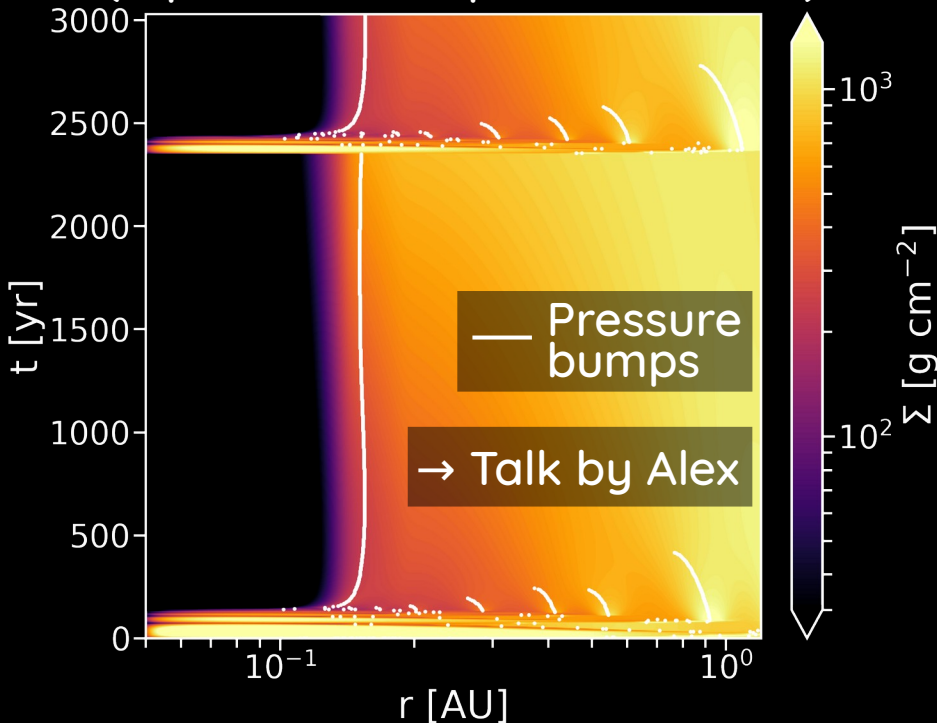


“Stationary” pressure maximum
at dead zone inner edge

Burst modes: Inner disk structure alteration

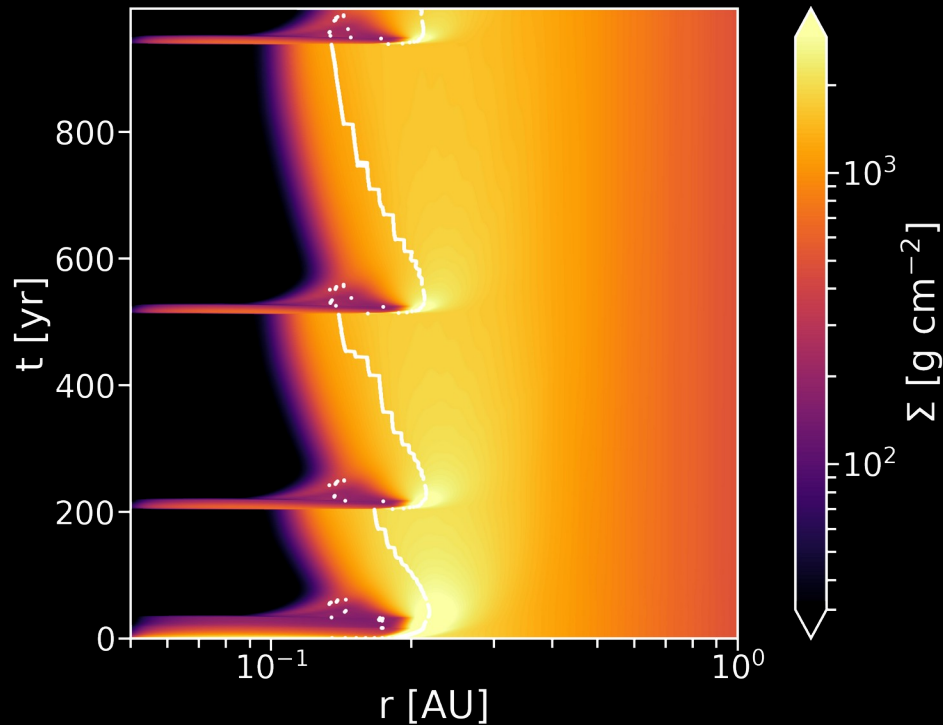
Wide burst mode

(equivalent “simple” T_{MRI} model)

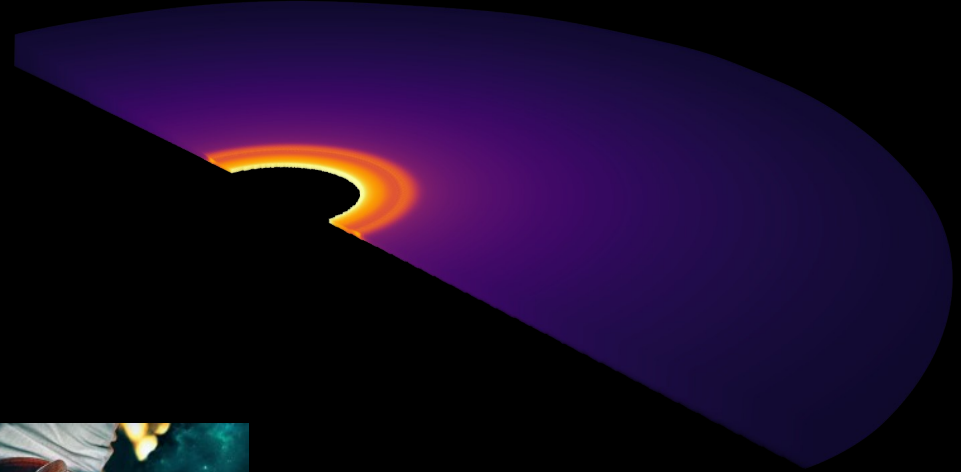
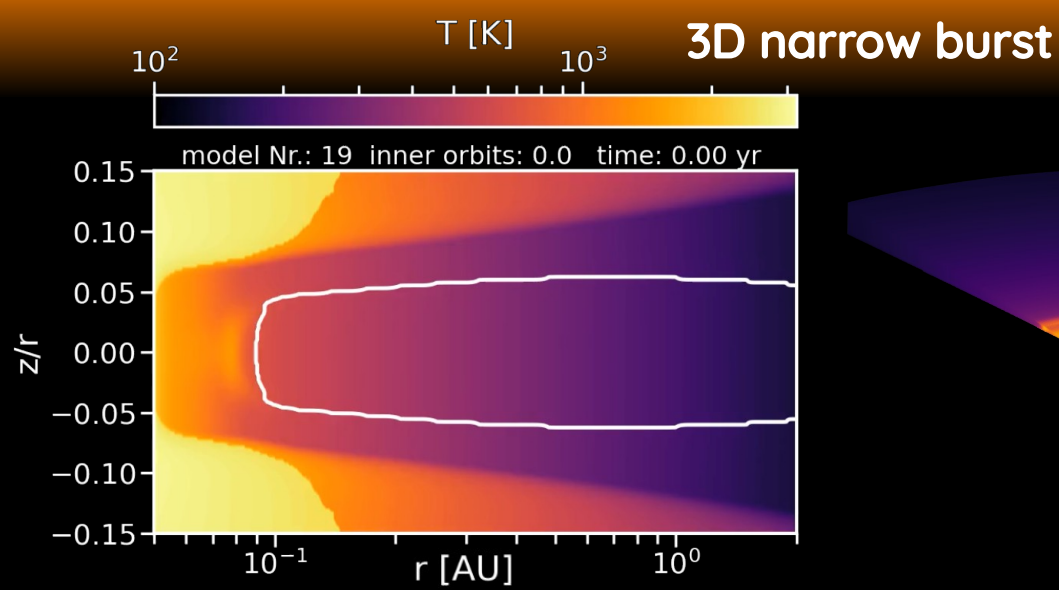


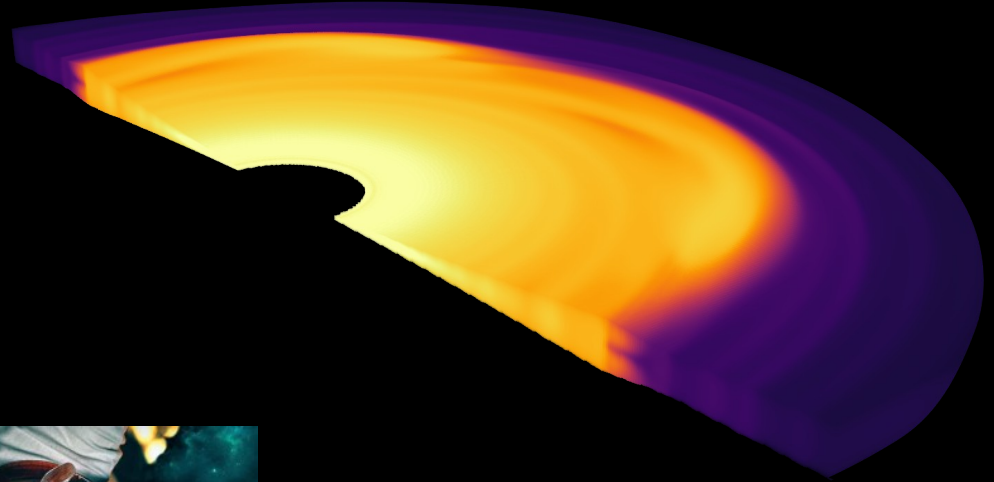
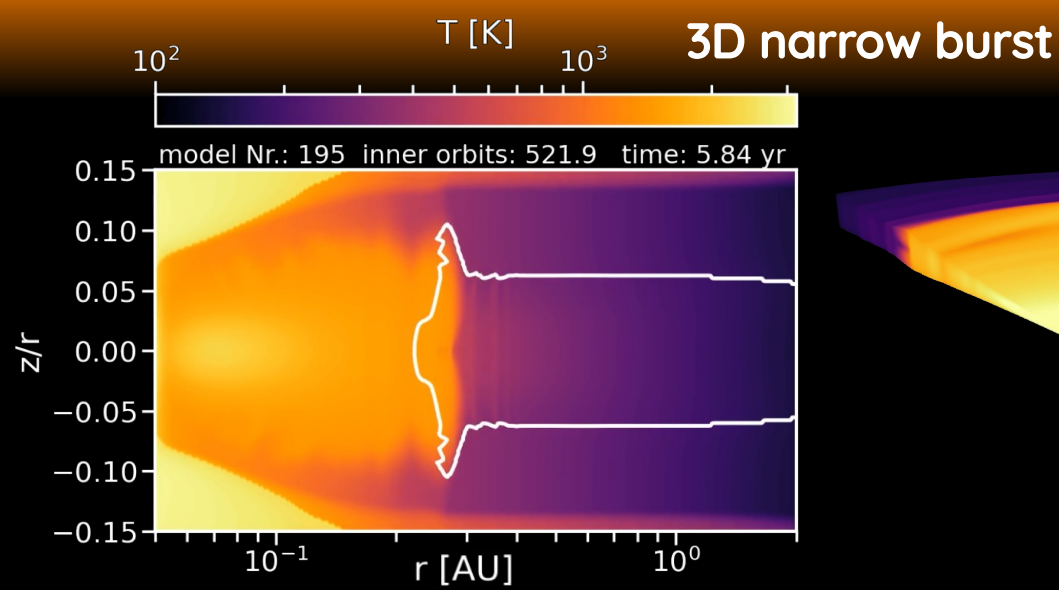
“Stationary” pressure maximum
at dead zone inner edge

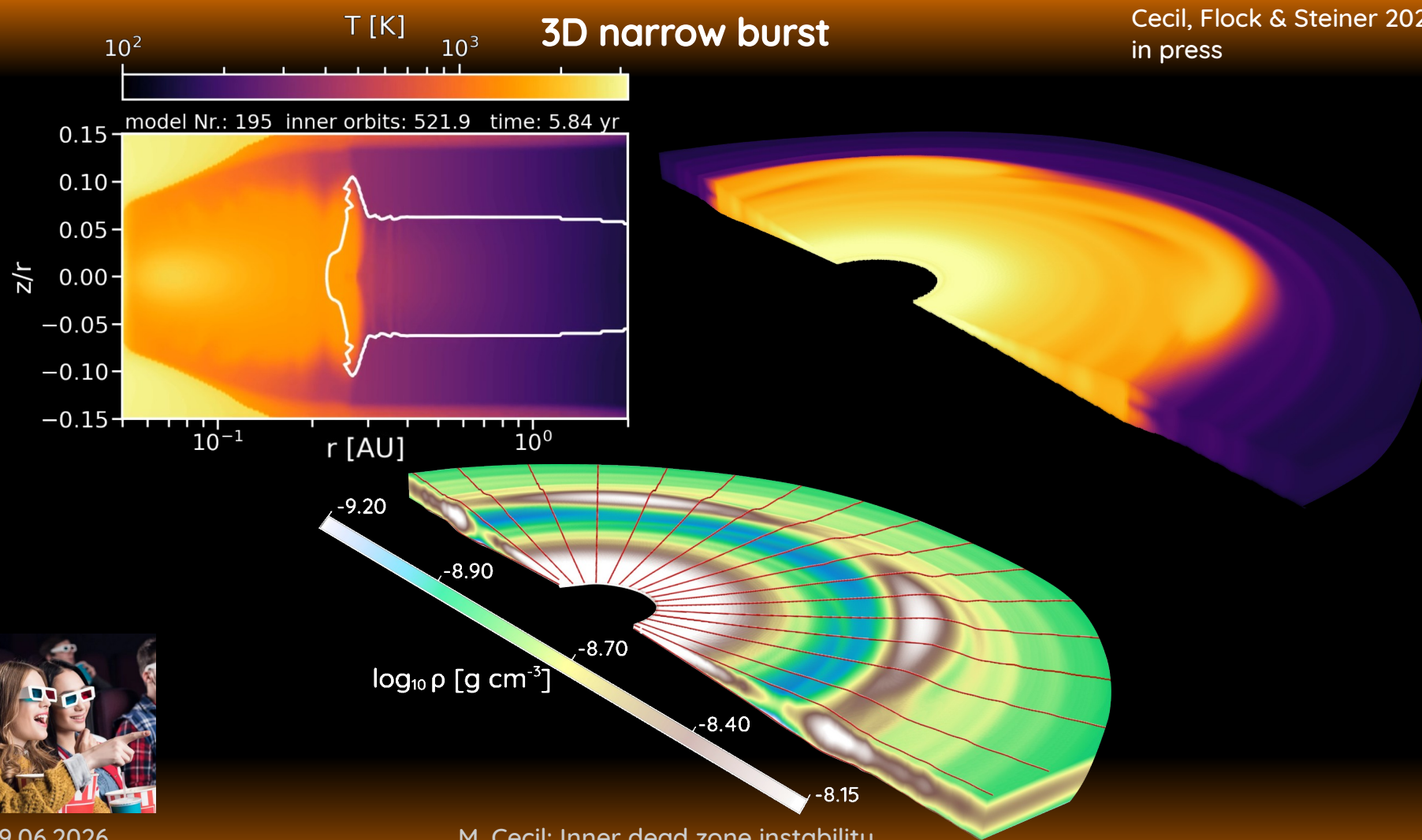
Narrow burst mode

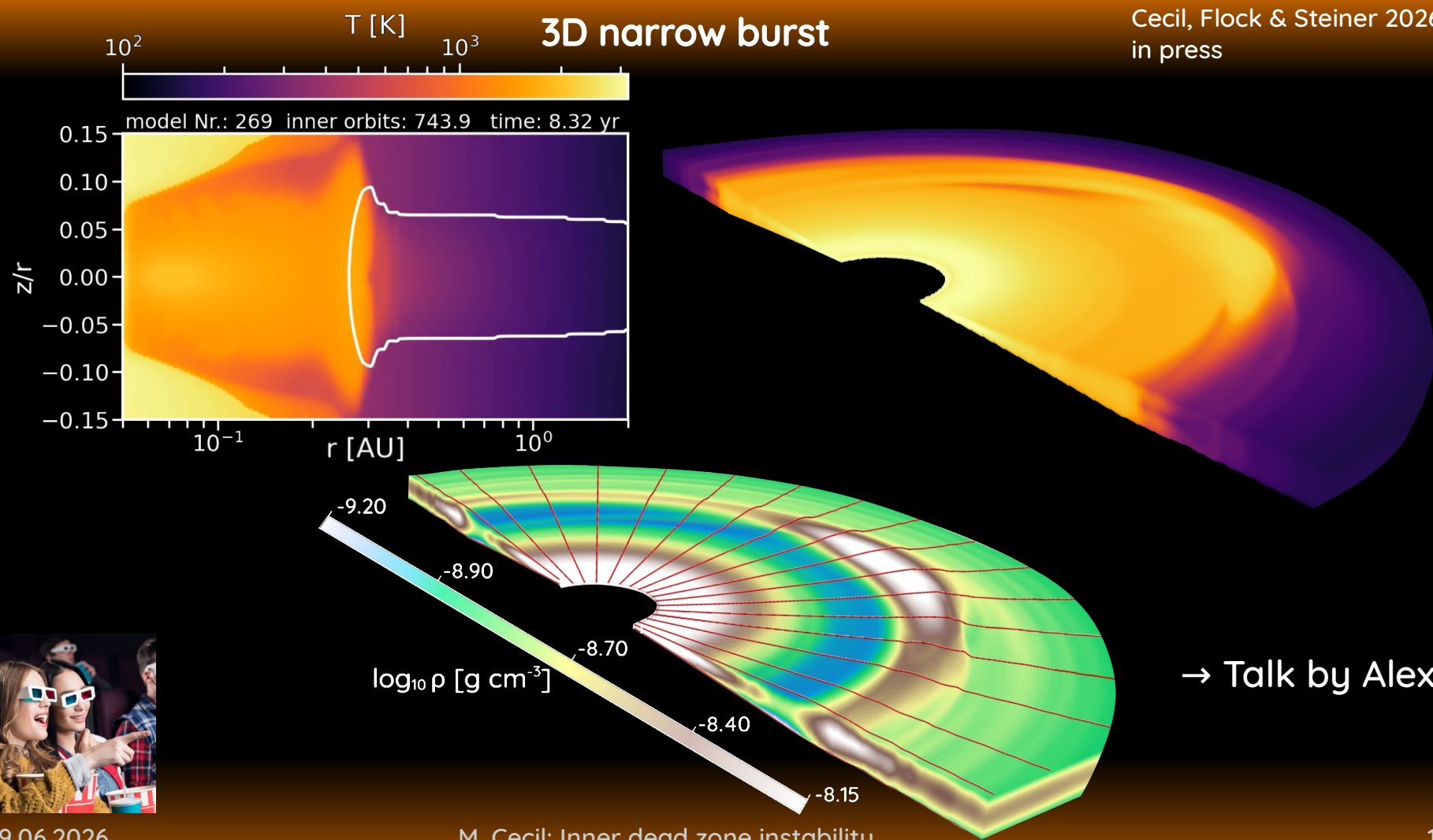


No stationary pressure maxima









RAVING WITH THE STARS:

Instability of the inner dead zone edge

Episode I (Cecil & Flock, 2024):

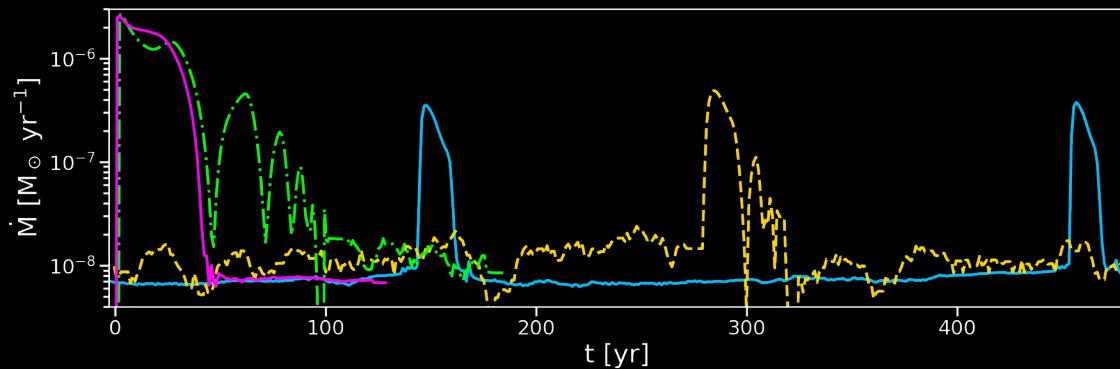
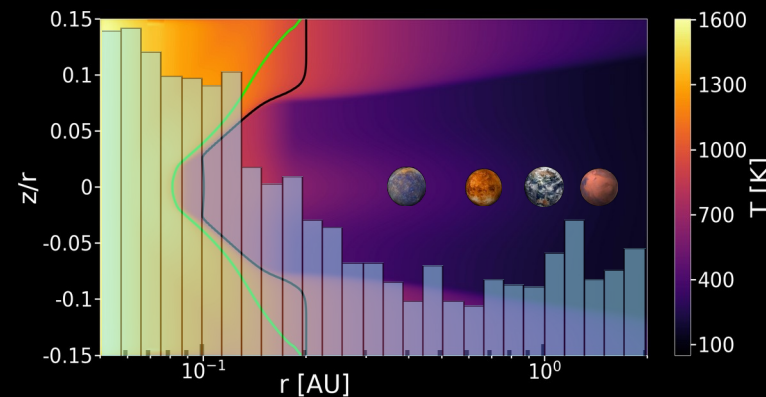
The inner dead zone is intrinsically unstable, leading to periodic accretion outbursts with possibly major consequences for planet formation and migration.

Episode II (Cecil et al. 2026a):

(Gas) opacities shift MRI and dust sublimation fronts but don't change outburst behaviour

Episode III (Cecil, Flock & Steiner, 2026b, in press):

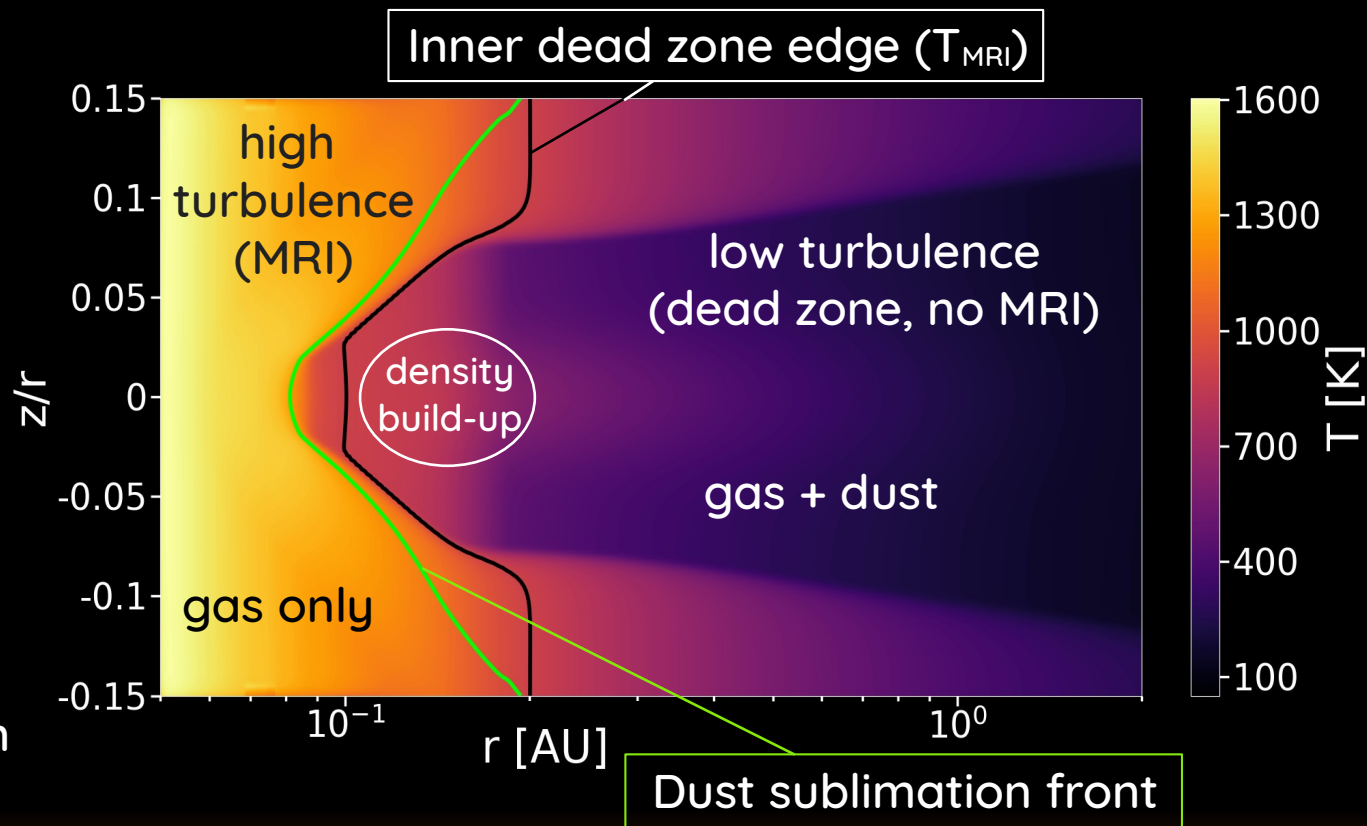
Burst morphology is highly sensitive to magnetic field strength, establishing distinct burst modes, each with different accretion signatures and impacts on planet-forming conditions



Backup

Radiation Hydrodynamics: Inner disk structure

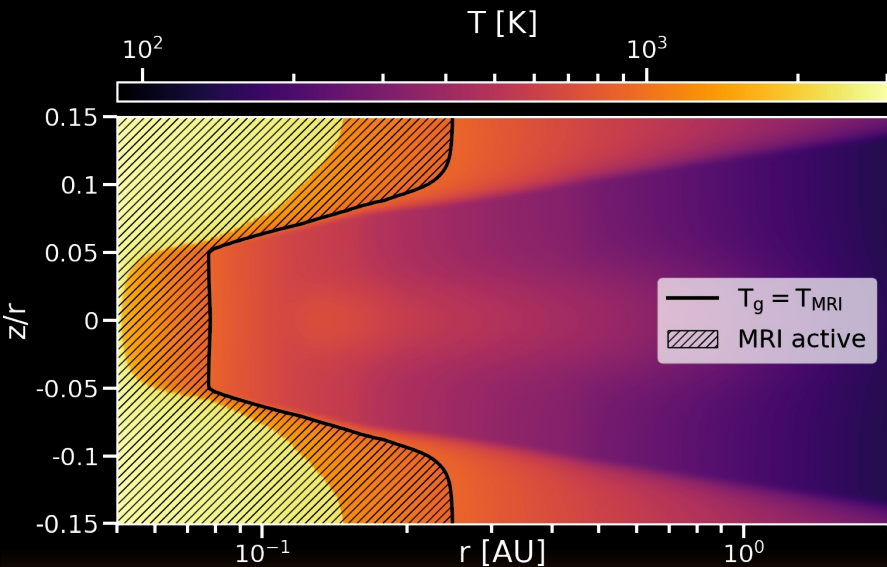
- Stellar irradiation
- Turbulent heating
- Radiative cooling
- Dust sublimation
- Turbulence transition (MRI)



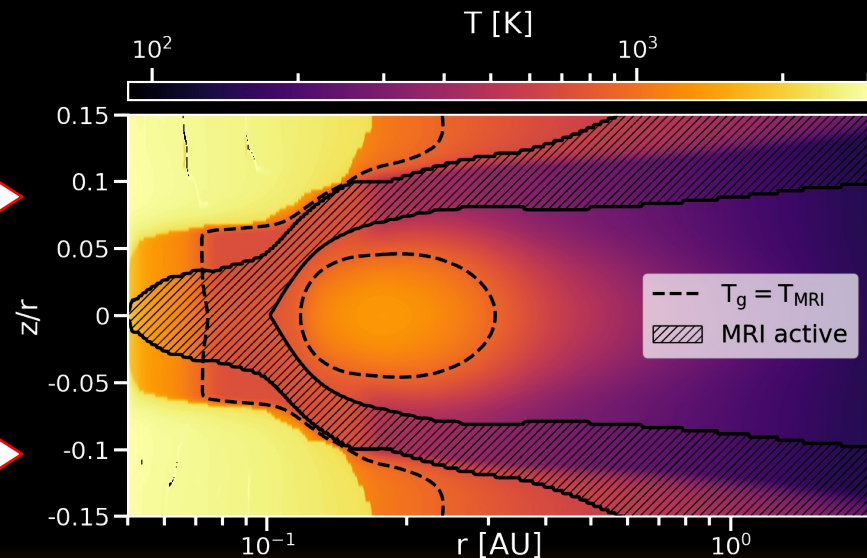
Is this stable, especially for low mass disks?

MRI activity tied to magnetic fields

T_{MRI} activation criterion



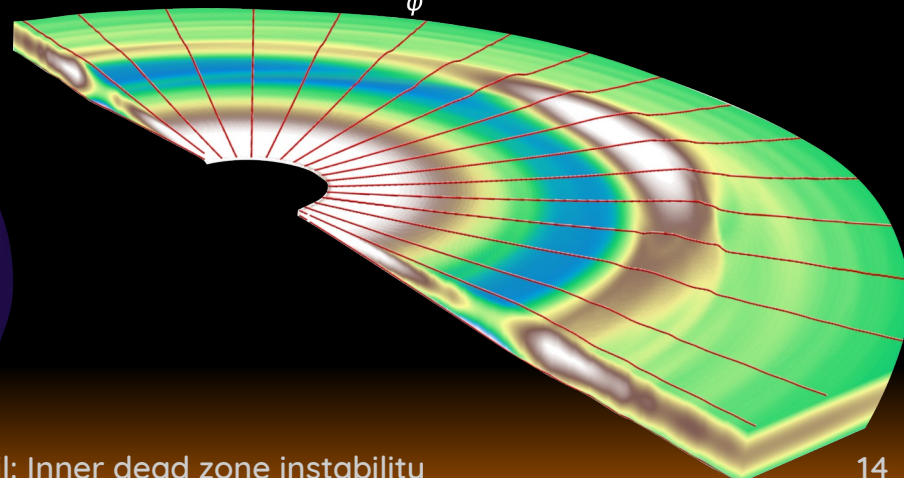
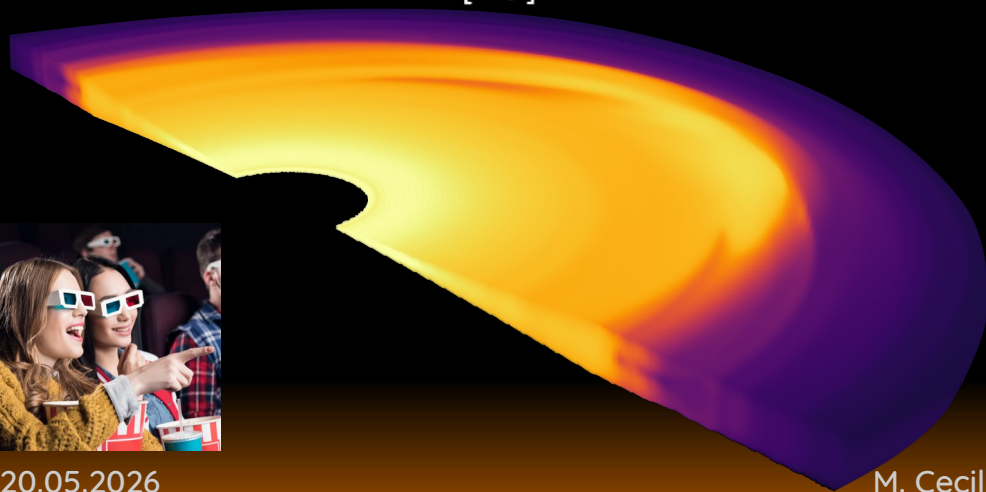
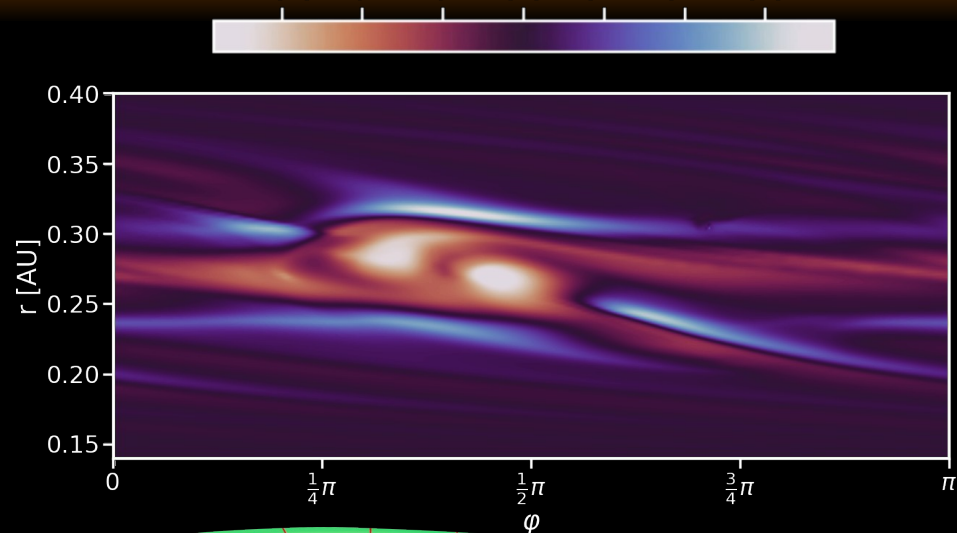
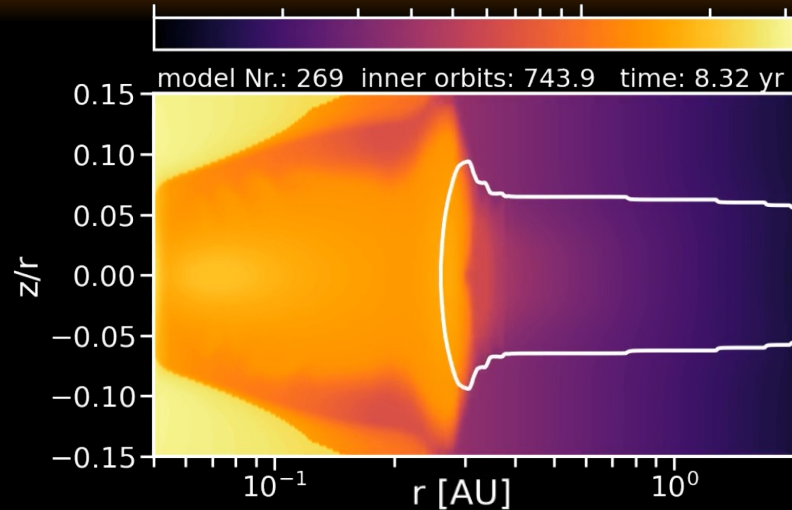
MRI activity determined by
magnetic field strength,
(stellar + disk component)



T [K]

3D narrow burst

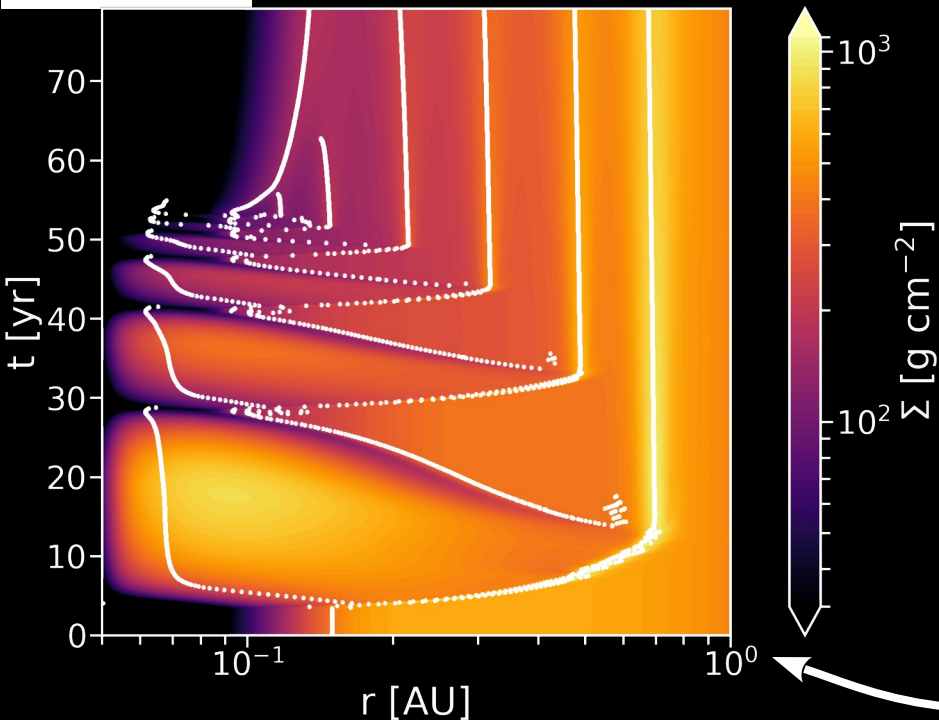
$(q - q_K)/q_K$



Episode I: Inner disk structure alteration

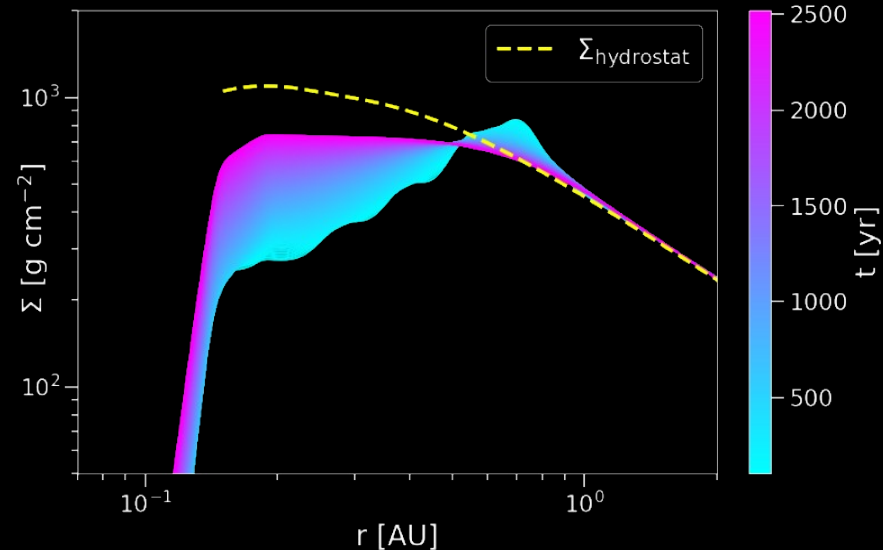


Instability phase



— Pressure maxima = dust traps

Quiescent phase



Density structure is (periodically) rearranged by instability



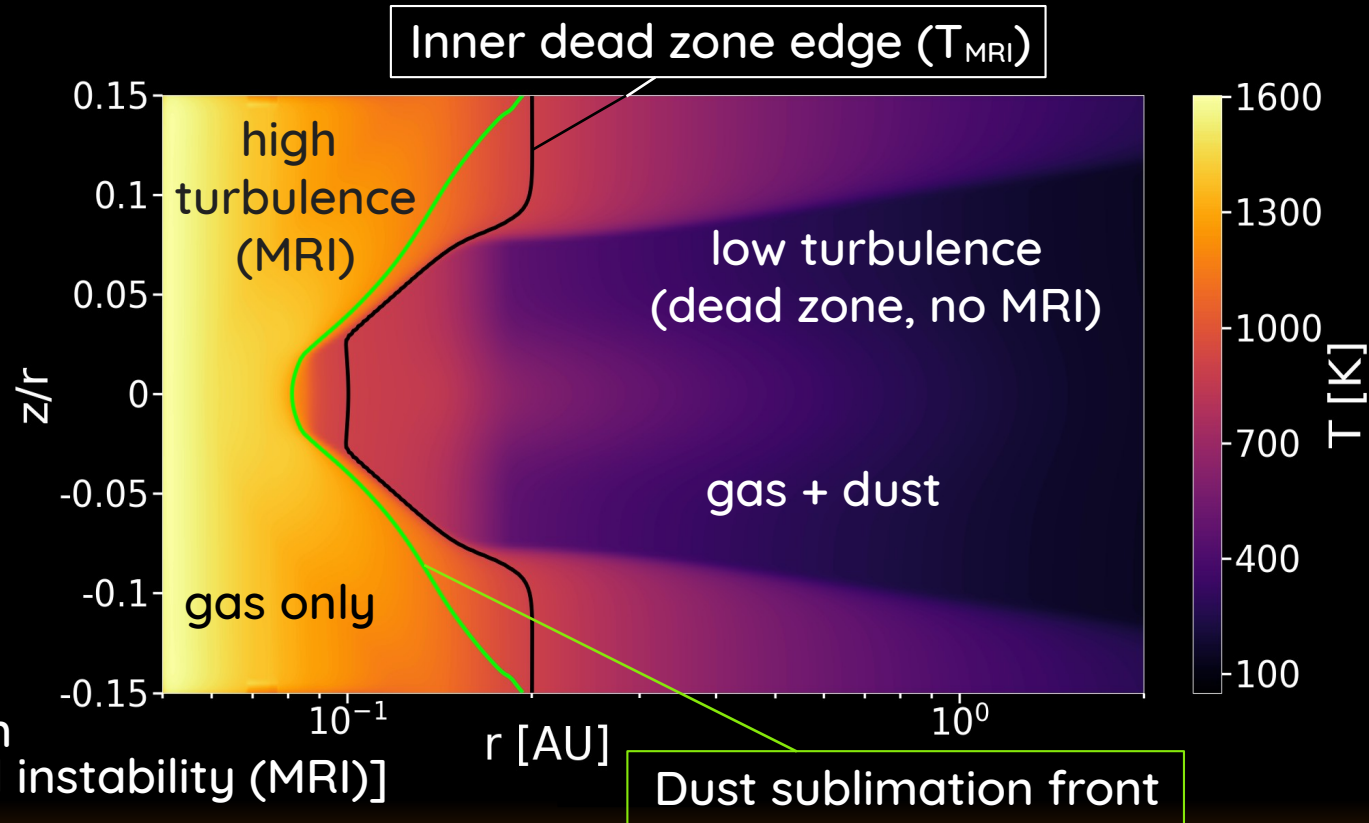
Radiation Hydrodynamics: Inner disk structure

Governed by
opacities

- Stellar irradiation
- Turbulent heating
- Radiative cooling

→ Dust sublimation

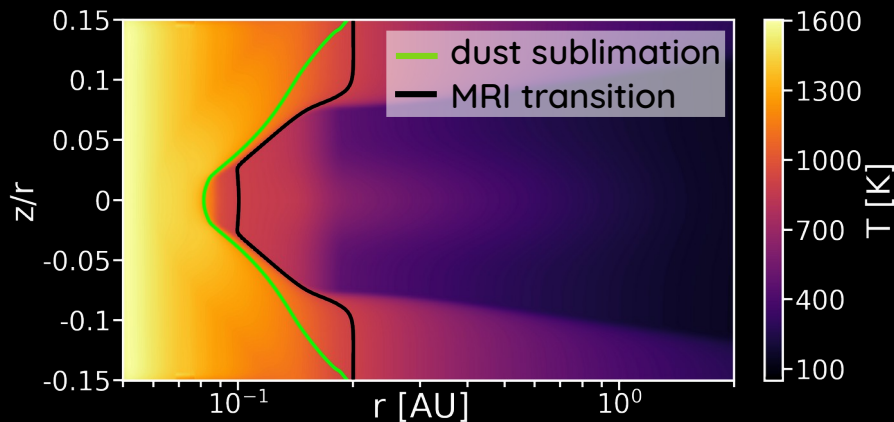
→ Turbulence transition
[magneto-rotational instability (MRI)]



Episode II: Opacity models

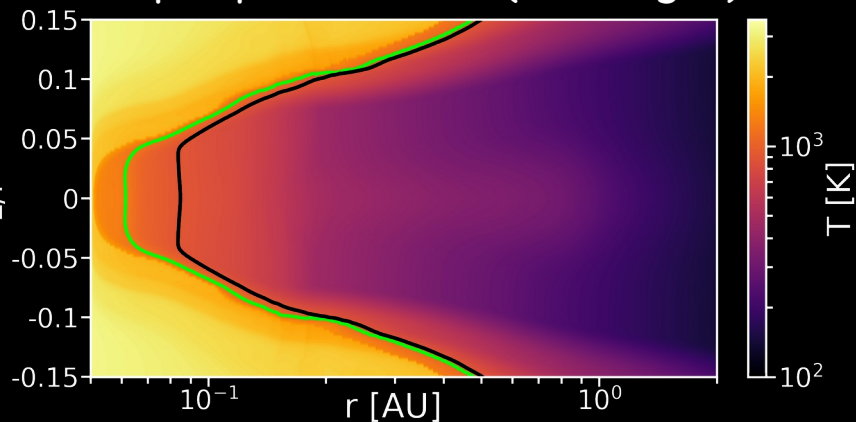
“simple”:

- $\kappa_P = \kappa_R = \text{const.}$ for gas and dust
- $\kappa_{\text{irr}} = \text{const.}$ (grey irradiation)



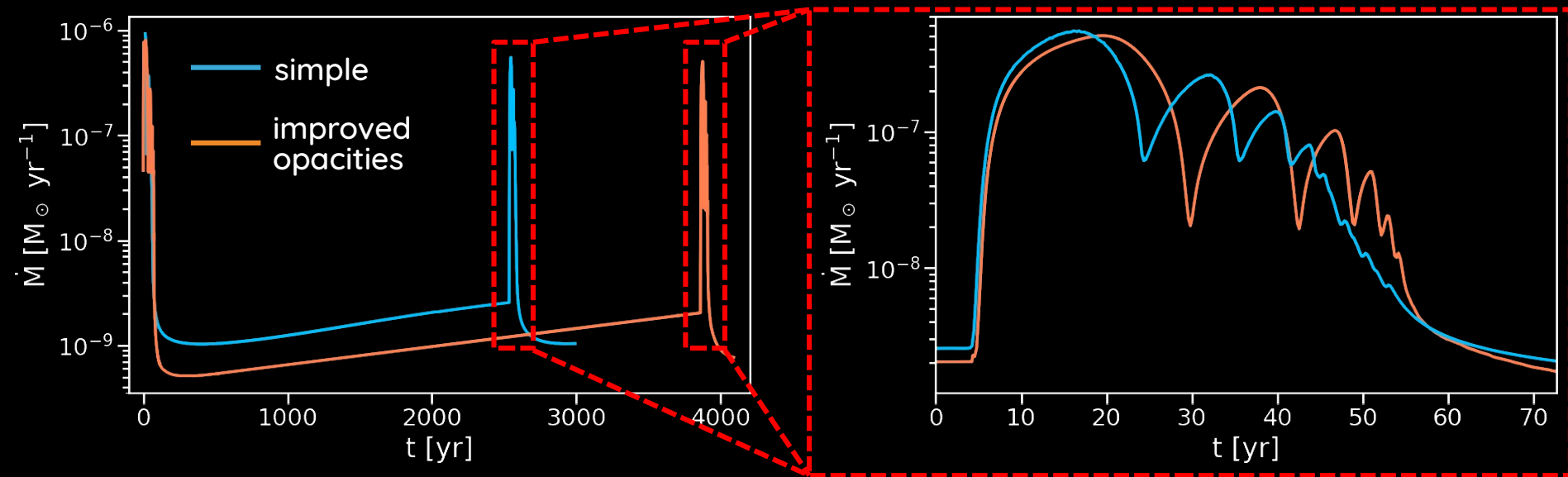
Improved:

- $\kappa_P, \kappa_R, \kappa_V$ for dust:
DIANA (Woitke et al. 2016)
- $\kappa_P, \kappa_R, \kappa_V$ for gas:
Malygin et al. 2014
- freq. dep. irradiation (dust + gas)



Gas opacities (primarily) have a strong effect on inner disk structure
However...

Episode II: Accretion rate modulation

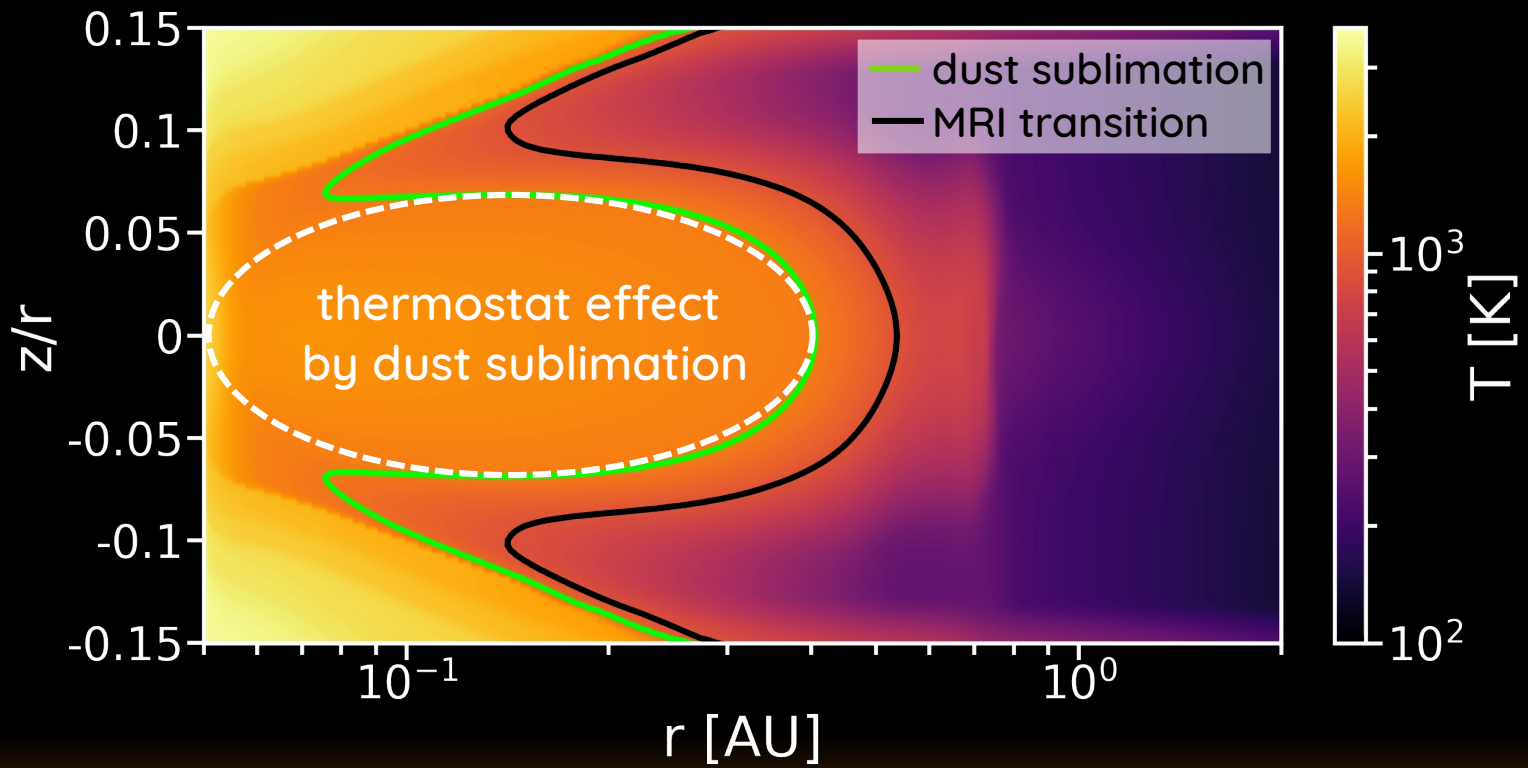


Opacities only affect quiescent timescale

Burst morphology remains robust

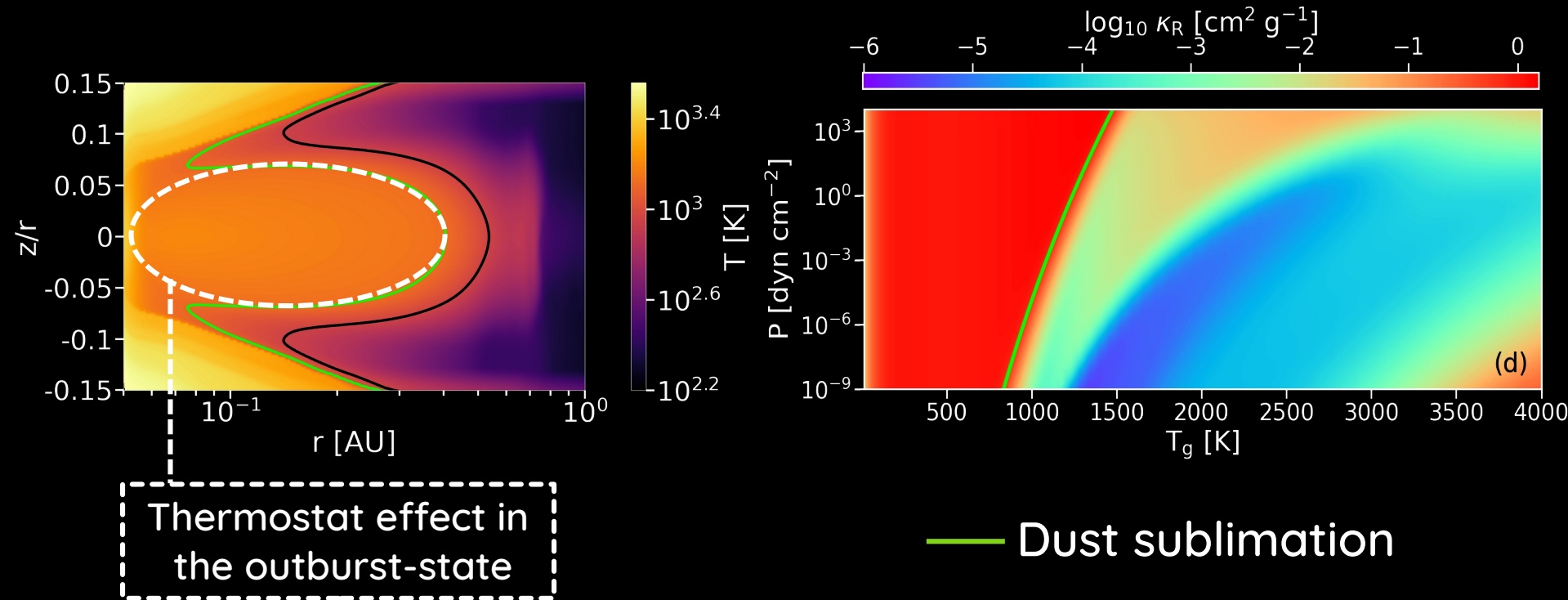
→ Findings on density and pressure evolution of Episode I apply equally

Episode II: Outburst state



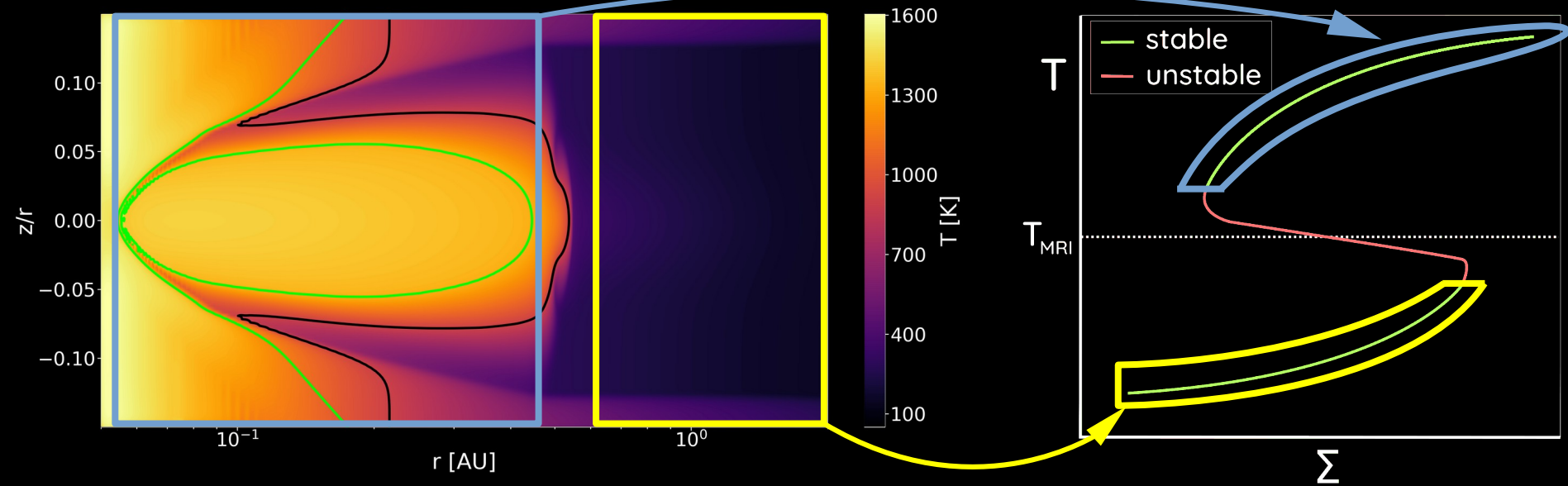
Gas opacities never dominant

Outburst state: Rosseland mean opacities



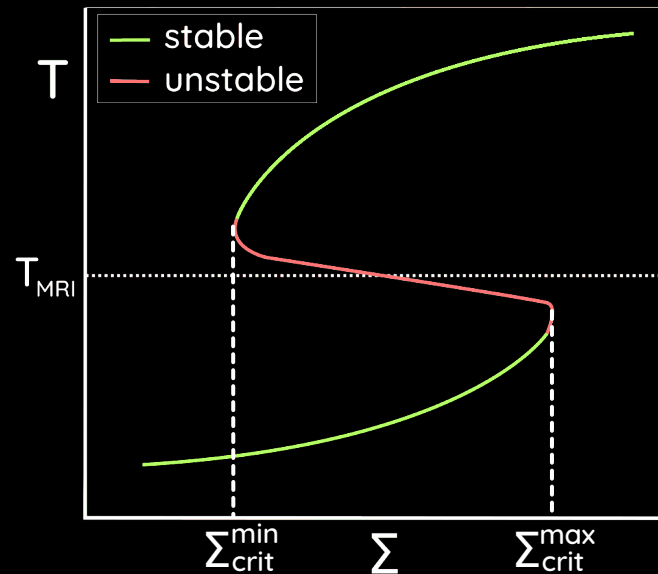
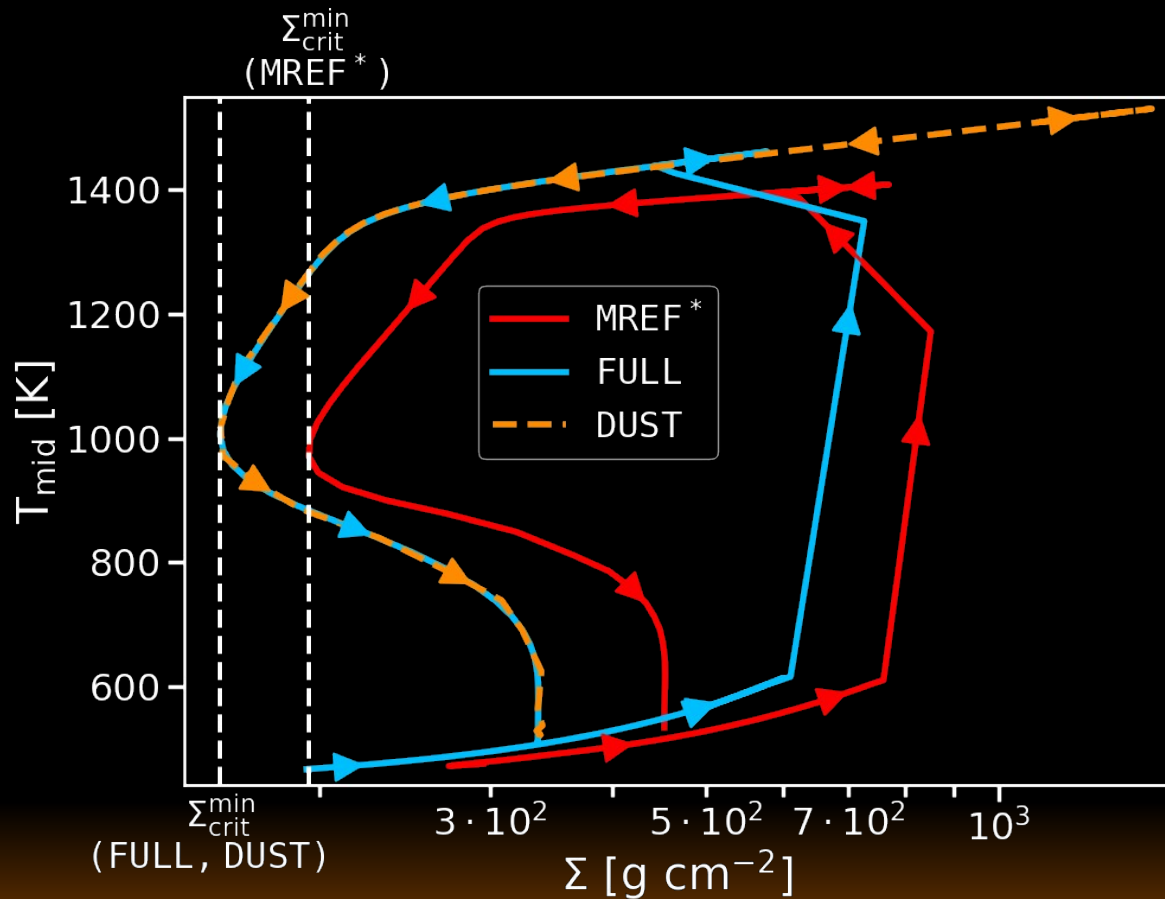
Tracking on S-curves of thermal equilibrium

‘High-state’ = high α



‘Low-state’ = low α

Tracking on S-curves of thermal equilibrium



“simple” RHD

$$T_g + T_{\text{MRI}}$$



$$\text{MRI}(T_g)$$

$$\alpha_{\text{max}} = \alpha_{\text{MRI}}$$

$$\alpha_{\text{min}} = \alpha_{\text{DZ}}$$

pseudo-non-ideal M-RHD

$$T_g + \rho_g + \zeta + \beta(P_g, B)$$



Ambipolar & Ohmic diffusivities
(η_A, η_o)
(Desch & Turner, 2015)



$$\text{MRI}(\eta_A, \eta_o, \rho_g, B, \Omega)$$

$$\alpha_{\text{max}} = \alpha_{\text{MRI}}$$

$$\alpha_{\text{min}} = \alpha_{\text{DZ}}$$